

Dysphagia Therapy - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses dysphagia therapy.

1. Medical Necessity

Aetna considers the following interventions medically necessary for treatment of dysphagia when criteria are met:

1. Speech therapy for treatment of dysphagia, regardless of the presence of a communication disability, for members who meet the criteria set forth below:
 1. Member exhibits weight loss or malnutrition because he/she has dysphagia and is unable to obtain adequate nutrition orally; *or*
 2. Member has a history of, or is at high-risk for, recurrent aspirations or choking; *or*
 3. Member is unable to swallow and has a nasogastric or gastrostomy tube for nutrition;
2. Esophageal dilation for the treatment of symptomatic obstruction of the esophagus;
3. Non-biodegradable, non-drug eluting stent placement for refractory (i.e., cannot be dilated to an adequate diameter) malignant esophageal strictures;
4. Per-oral endoscopic myotomy (POEM) for the treatment of type III (spastic) achalasia.

2. Experimental, Investigational, or Unproven

Aetna considers the following interventions (not an all-inclusive list) experimental, investigational, or unproven for treatment of dysphagia because the effectiveness of these approaches has not been established:

1. Acupuncture
2. Artificial intelligence-based technology for the management of dysphagia
3. Biodegradable stent

4. Drug-eluting stents for esophageal strictures (malignant or benign)
5. Dysphagia therapy for all other indications other than the ones listed in Section I
6. Electrical stimulation (including neuromuscular electrostimulation, and pharyngeal electrical stimulation)
7. ERBE electrocautery
8. Esophageal dilation for the treatment of non-obstructive esophageal dysphagia
9. Hyperbaric oxygen therapy for the treatment of radiation-induced dysphagia
10. Intensive dysphagia rehabilitation approach/program (e.g., the Swallow STRONG multi-disciplinary oropharyngeal strengthening program)
11. Laryngeal manipulation
12. Mirror therapy
13. Per oral plication of the esophagus (POPE) for the treatment of dysphagia
14. Pharyngeal motor cortex stimulation
15. POEM for achalasia (other than type III achalasia)
16. Repetitive peripheral/transcranial magnetic stimulation
17. Transcranial direct current stimulation.

3. Policy Limitations and Exclusions

Some plans limit coverage of medically necessary speech therapy services. Members should check their benefit plan descriptions for any applicable benefit plan limitations and exclusions on coverage for speech therapy services.

4. Related Policies

- [CPB 0113 - Botulinum Toxin](#) - for botulinum toxin for treatment of dysphagia due to achlasia
- [CPB 0135 - Acupuncture](#)
- [CPB 0243 - Speech Therapy](#)
- [CPB 0621 - Drug-Eluting Stents](#)

CPT Codes / ICD-10 Codes / HCPCS Codes

Code	Code Description
<i>Therapy for the management of dysphagia other than esophageal dilation or stent placement:</i>	
CPT codes covered if selection criteria are met:	
43497	Lower esophageal myotomy, transoral (ie, peroral endoscopic myotomy [POEM])
92507	Treatment of speech, language, voice, communication, and/or auditory processing disorder; individual [not covered for intensive dysphagia rehabilitation program]

Code	Code Description
92508	Treatment of speech, language, voice, communication, and/or auditory processing disorder; group, 2 or more individuals [not covered for intensive dysphagia rehabilitation program]
92526	Treatment of swallowing dysfunction and/or oral function for feeding
92610	Evaluation of oral and pharyngeal swallowing function
92611	Motion fluoroscopic evaluation of swallowing function by cine or video recording
92612	Flexible fiberoptic endoscopic evaluation of swallowing by cine or video recording
92613	physician interpretation and report only
CPT codes not covered for indications listed in the CPB:	
<i>Repetitive peripheral magnetic stimulation, Swallow STRONG multi-disciplinary oropharyngeal strengthening program, pharyngeal electrical stimulation and transcranial direct current stimulation, mirror therapy, per oral plication of the esophagus (POPE), artificial intelligence-based technology - no specific code</i>	
43229	Esophagoscopy, flexible, transoral; with ablation of tumor(s), polyp(s), or other lesion(s) (includes pre- and post-dilation and guide wire passage, when performed) [not covered for ERBE electrocautery]
90867	Therapeutic repetitive transcranial magnetic stimulation (TMS) treatment; initial, including cortical mapping, motor threshold determination, delivery and management
90868	subsequent delivery and management, per session
90869	subsequent motor threshold re-determination with delivery and management
97014	Application of a modality to one or more areas; electrical stimulation (unattended)
97032	Application of a modality to one or more areas; electrical stimulation (manual), each 15 minutes
97810 - 97814	Acupuncture
99183	Physician or other qualified health care professional attendance and supervision of hyperbaric oxygen therapy, per session
HCPCS codes covered if selection criteria are met:	
G0153	Services performed by a qualified speech-language pathologist in the home health or hospice setting, each 15 minutes
S9128	Speech therapy, in the home, per diem
HCPCS codes not covered for indications listed in the CPB:	
A4575	Topical hyperbaric oxygen chamber, disposable

Code	Code Description
E0446	Topical oxygen delivery system, not otherwise specified, includes all supplies and accessories
E0720	Transcutaneous electrical nerve stimulation (TENS) device, two lead, localized stimulation
E0730	Transcutaneous electrical nerve stimulation (TENS) device, four or more leads, for multiple nerve stimulation
E0745	Neuromuscular stimulator, electronic shock unit
G0277	Hyperbaric oxygen under pressure, full body chamber, per 30 minute interval
G0283	Electrical stimulation (unattended), to one or more areas for indication(s) other than wound care, as part of a therapy plan of care
ICD-10 codes covered if selection criteria are met:	
• I69.091 • I69.191 • I69.291 • I69.391 • I69.891 • I69.991	Cerebrovascular disease, dysphagia, sequelae
K22.0	Achalasia of cardia [Type III achalasia]
R13.10 - R13.19	Dysphagia
ICD-10 codes not covered for indications listed in the CPB:	
T66.XXXA - T66.XXXS	Radiation sickness, unspecified
<i>Esophageal dilation:</i>	
CPT codes covered if selection criteria are met:	
43196	Esophagoscopy, rigid, transoral; with insertion of guide wire followed by dilation over guide wire
43213	Esophagoscopy, flexible, transoral; with dilation of esophagus, by balloon or dilator, retrograde (includes fluoroscopic guidance, when performed)
43214	with dilation of esophagus with balloon (30 mm diameter or larger) (includes fluoroscopic guidance, when performed)
43220	Esophagoscopy, rigid or flexible; with balloon dilation (less than 30 mm diameter)

Code	Code Description
43226	with insertion of guide wire followed by dilation over guide wire
43229	Esophagoscopy, flexible, transoral; with ablation of tumor(s), polyp(s), or other lesion(s) (includes pre- and post-dilation and guide wire passage, when performed)
43233	Esophagogastroduodenoscopy, flexible, transoral; with dilation of esophagus with balloon (30 mm diameter or larger) (includes fluoroscopic guidance, when performed)
43249	Upper gastrointestinal endoscopy including esophagus, stomach, and either the duodenum and/or jejunum as appropriate; with balloon dilation of esophagus (less than 30 mm diameter)
43253	Esophagogastroduodenoscopy, flexible, transoral; with transendoscopic ultrasound-guided transmural injection of diagnostic or therapeutic substance(s) (eg, anesthetic, neurolytic agent) or fiducial marker(s) (includes endoscopic ultrasound examination of the esophagus, stomach, and either the duodenum or a surgically altered stomach where the jejunum is examined distal to the anastomosis)
43270	Esophagogastroduodenoscopy, flexible, transoral; with ablation of tumor(s), polyp(s), or other lesion(s) (includes pre- and post-dilation and guide wire passage, when performed)
43450	Dilation of esophagus, by unguided sound or bougie, single or multiple passes
43453	Dilation of esophagus, over guide wire
ICD-10 codes covered if selection criteria are met:	
K22.2	Esophageal obstruction
• Q39.0 - Q39.4 • Q39.8	Congenital malformations of esophagus
ICD-10 codes not covered for indications listed in the CPB:	
• I69.091 • I69.191 • I69.291 • I69.391 • I69.891 • I69.991	Cerebrovascular disease, dysphagia, sequelae
R13.10 - R13.19	Dysphagia
Stent Placement:	

Code	Code Description
CPT codes covered if selection criteria are met:	
43212	Esophagoscopy, flexible, transoral; with placement of endoscopic stent (includes pre- and post-dilation and guide wire passage, when performed)
43266	Esophagogastroduodenoscopy, flexible, transoral; with placement of endoscopic stent (includes pre- and post-dilation and guide wire passage, when performed)
HCPCS codes covered if selection criteria are met:	
C1874	Stent, coated/covered, with delivery system [not covered for biodegradable and drug-eluting stent]
C1875	Stent, coated/covered, without delivery system [not covered for biodegradable and drug-eluting stent]
Other HCPCS codes related to the CPB:	
C1876	Stent, noncoated/noncovered, with delivery system
C1877	Stent, noncoated/noncovered, without delivery system
ICD-10 codes covered if selection criteria are met:	
C15.3 - C15.9	Malignant neoplasm of esophagus
ICD-10 codes not covered for indications listed in the CPB:	
• I69.091, • I69.191, • I69.291, • I69.391, • I69.891, • I69.991	Sequelae of cerebrovascular disease, dysphagia
R13.10 - R13.19	Dysphagia

Background

People with dysphagia have difficulty swallowing and may also experience pain while swallowing. Some people may be completely unable to swallow or may have trouble swallowing liquids, foods, or saliva.

Dysphagia occurs when there is a problem with any part of the swallowing process. Weak tongue or cheek muscles may make it hard to move food around in the mouth for chewing. Food pieces that are too large for swallowing may enter the throat and block the passage of air.

Other problems include not being able to start the swallowing reflex (a stimulus that allows food and liquids to move safely through the pharynx) because of a stroke or other nervous system disorder. People with these kinds of problems are unable to begin the muscle movements that allow food to move from the mouth to the stomach. Another difficulty can occur when weak throat muscles cannot move all of the food toward the stomach. Bits of food can fall or be pulled into the trachea, which may result in aspiration pneumonia.

Dysphagia may be caused by any condition that weakens or damages the muscles and nerves used for swallowing. For example, people with nervous system diseases, such as cerebral palsy or Parkinson's disease, often have problems swallowing. Additionally, cerebrovascular accident or traumatic brain injury may affect the coordination of the swallowing muscles or limit sensation in the mouth and throat. An infection or irritation can cause narrowing of the esophagus. People born with abnormalities of the swallowing mechanism may not be able to swallow normally. Infants who are born with a cleft palate are unable to suck properly, which complicates breast-feeding and drinking from a regular baby bottle. In addition, cancers of the head, neck, or esophagus may cause dysphagia. Sometimes the treatment for these types of cancers can cause dysphagia. Injuries of the head, neck, and chest may also create swallowing problems.

Physicians and speech-language pathologists who test for and treat swallowing disorders use a variety of tests that allow them to look at the parts of the swallowing mechanism, including fiber optic laryngoscopy, video fluoroscopy, and ultrasound.

Once the cause of the dysphagia is found, surgery or medication may help. If treating the cause of the dysphagia does not help, the patient may refer the patient to a speech-language pathologist who is trained in testing and treating swallowing disorders. The speech-language pathologist will test the person's ability to eat and drink and may teach the person new ways to swallow.

Oral-motor therapy is directed at correcting abnormal oral muscle behaviors that interfere with feeding. Oral-motor therapy may be focused upon inducing active suckle movements, coordinating tongue movements, or facilitating normal oral movement patterns such as lip closure.

Oral-motor therapy has also been used in developmentally delayed children to stop drooling, correct abnormal tongue thrust, and improve speech. Speech management of the developmentally delayed child has included training to improve the functioning of oral and pharyngeal muscles. This oral-motor training is usually introduced before the emergence of speech. Most Aetna plans exclude treatment of developmental delay. Please check benefit plan descriptions for details.

Non-Biodegradable Stent for Malignant Esophageal Strictures

Verschuur and associates (2007) compared small- and large-diameter stents for improvement of dysphagia, complications, and recurrent dysphagia. These investigators prospectively followed 338 patients with dysphagia from obstructing esophageal or gastric cardia cancer who were treated with an Ultra-flex stent (n = 153), a Gianturco Z-stent (n = 89), or a Flamingo Wallstent (n = 96) of either a small diameter (n = 265) or a large diameter (n = 73). Major outcome measurements included dysphagia score (on a scale from 0 [no dysphagia] to 4 [complete dysphagia]), complications, and recurrent dysphagia. Analysis was by Chi-2 test, log-rank test, and Cox regression analysis. Improvement in dysphagia was similar between patients with a small- or a large-diameter stent (p = 0.35). The occurrence of major complications, such as hemorrhage, perforation, fistula, and fever, was increased in patients with a large-diameter Gianturco Z-stent compared with those treated with a small-diameter stent (4 [40 %] versus 16 [20 %]; adjusted hazard ratio [HR] 5.03, 95 % confidence interval [CI]: 1.33 to 19.11) but not in patients with a large-diameter Ultra-flex stent or a Flamingo Wallstent. Moreover, minor complications, particularly pain, were associated with prior radiation and/or chemotherapy in patients with a large- or a small-diameter Gianturco Z-stent (HR 4.27, 95 % CI: 1.44 to 12.71) but not in those with an Ultra-flex stent or a Flamingo Wallstent. Dysphagia from stent migration, tissue overgrowth, and food bolus obstruction recurred more frequently in patients with a small-diameter stent than in those with a large-diameter stent (Ultra-flex stent: 54 [42 %] versus 3 [13 %], adjusted HR 0.16, 95 % CI: 0.04 to 0.74; Gianturco Z-stent: 21 [27 %] versus 1 [10 %], adjusted HR 0.97, 95 % CI: 0.11 to 8.67; and Flamingo Wallstent: 21 [37 %] versus 6 [15 %], adjusted HR 0.40, 95 % CI: 0.03 to 4.79). The authors concluded that large-diameter stents reduce the risk of recurrent dysphagia from stent migration, tissue overgrowth, or food obstruction. Increasing the diameter in some

stent types may, however, increase the risk of stent-related complications to the esophagus.

Conio et al (2007) stated that self-expanding metal stents (SEMS) provide effective palliation in patients with malignant dysphagia, although severe complications and mortality may result. These researchers performed a prospective controlled trial to compare a new self-expanding polyester mesh stent (Polyflex) with SEMS (Ultra-flex). A total of 101 patients with unresectable esophageal carcinoma were randomized to placement of a Polyflex (n = 47) or a partially covered Ultra-flex (n = 54) stent. Patients with esophagogastric junction malignancy were excluded. Placement was successful in 46 (98 %) patients with the Polyflex and 54 (100 %) patients with the Ultraflex stent. In 1 patient, the Polyflex stent could not be placed. After 1 week, dysphagia was improved by at least 1 grade in 100 % of the Polyflex group and in 94 % of the Ultraflex group. Major complications were observed in 48 % of the Polyflex group and 33 % of the Ultraflex group. Intra-procedural perforation occurred in 1 Polyflex and 1 Ultraflex patient; 2 Polyflex patients had post-procedural hemorrhage. Twenty (44 %) patients with a Polyflex stent and 18 (33 %) with an Ultraflex stent had recurrent dysphagia because of tumor overgrowth, stent migration, hyperplastic granulomatous reaction, or food bolus impaction. Multi-variate analyses showed a significantly higher complication rate with Polyflex than with Ultraflex stents (odds ratio 2.3, 95 % CI: 1.2 to 4.4). However, median survival was 134 days with Polyflex and 122 days with Ultraflex stents (p = NS). The authors concluded that no difference was seen in palliation of dysphagia between the 2 stents. Significantly more complications, especially late stent migration, were observed in the Polyflex group.

Verschuur and colleagues (2008) noted that stents are often used for the palliation of inoperable esophageal or gastric cardia cancer. One of the drawbacks of the currently used stents is the high percentage of recurrent dysphagia due to stent migration and tissue growth. New stents have been designed to overcome this unwanted sequela of stent placement. In this randomized trial, these investigators examined if results of stent placement could be improved with newer stent designs. A total of 125 patients with dysphagia from inoperable carcinoma of the esophagus or gastric cardia were randomized to placement of an Ultraflex stent (n = 42), Polyflex stent (n = 41), or Niti-S stent (n = 42). Patients were followed by scheduled telephone calls at 14 days after treatment, and then monthly for 6 months or until death. Technical and functional outcome, complications, recurrent dysphagia, and survival were analyzed with Chi(2) tests, Kaplan-Meier curves, and log-rank tests. Stent placement was technically successful in all patients with an Ultraflex stent, in 34/41 (83 %) patients with a Polyflex stent, and in 40/42 (95 %) patients treated with a Niti-S stent (p = 0.008). Dysphagia score improved from a median of 3 (liquids only) to 1 (ability to eat some solid food) in all patients. There were no differences in complications among the 3 stent types. Recurrent dysphagia, caused by tissue ingrowth or overgrowth, migration, or food obstruction, was significantly different between patients with an Ultraflex stent and patients with a Polyflex stent or Niti-S stent (22 [52 %] versus 15 [37 %] versus 13 [31 %], p = 0.03). Stent migration occurred more frequently with Polyflex stents, whereas tissue ingrowth or overgrowth was more frequently seen with Ultraflex stents, and to a lesser degree, Niti-S stents. No differences were found in survival (median survival: Ultraflex stent 132 days versus Polyflex stent 102 days versus Niti-S stent 159 days) among the 3 stent types. The authors concluded that all 3 stents are safe and offer adequate palliation of dysphagia from esophageal or gastric cardia cancer. Nonetheless, Polyflex stents seem the least preferable in this patient group, as placement of this device is technically demanding and associated with a high rate of stent migrations.

Esophageal Dilation

Guidelines on the use of esophageal dilation (Riley and Attwood, 2004) stated that esophageal dilation is indicated in the treatment of symptomatic obstruction of the esophagus. The guidelines explained that obstruction may develop as a consequence of a wide range of anatomical and functional esophageal disorders. Reflux-induced strictures, malignant strictures, and achalasia are the most frequent indications but patients with anastomotic, sclerotherapy, radiation, medication, and corrosive induced strictures, and those with rings and webs frequently require dilatation. The guidelines stated that patients with diffuse esophageal spasm and other motility disorders may occasionally require dilatation of the lower esophageal sphincter when conservative measures fail.

There is inadequate evidence of the clinical utility of esophageal dilation in dysphagia not associated with obstruction. In a randomized controlled trial (n = 96), Lavu et al

(2004) examined the impact of esophageal dilation with a large-diameter dilator on dysphagia and quality of life in such patients. These investigators found that most patients with esophageal dysphagia have a non-obstructing esophageal lumen. Their findings did not support the practice of empiric esophageal dilation for patients with non-obstructive esophageal dysphagia. Improvement in both treatment and control groups suggests that it occurred due to proton pump inhibitor therapy, lending credence to the hypothesis that esophageal hypersensitivity to acid contributes to symptoms in most patients with non-obstructive esophageal dysphagia, which is the predominant category of dysphagia.

Siersema (2008) stated that esophageal strictures are frequently encountered by gastroenterologists and can be caused by benign or malignant lesions. Dysphagia is the symptom experienced by all patients, regardless of the cause of their strictures. The methods most commonly employed for palliation of malignant esophageal strictures are stent placement (particularly in patients with an expected survival of 3 months or less) and brachytherapy (in patients with a life expectancy of more than 3 months). Brachytherapy has been shown to be beneficial in patients with an expected survival of longer than 3 months with regard to (prolonged) dysphagia improvement, complications and quality of life. The mainstay of benign esophageal stricture treatment is dilation. Although dilation usually results in symptomatic relief, strictures do recur. In order to predict which types of strictures are most likely to recur, it is important to differentiate between esophageal strictures that are simple (i.e., focal, straight strictures with a diameter that allows endoscope passage) and those that are more complex (i.e., over 2 cm in length, tortuous strictures with a narrow diameter). These complex strictures are considered refractory when they can not be dilated to an adequate diameter.

Acupuncture

In a meta-analysis, Long and Wu (2012) examined the effect of acupuncture for treatment of dysphagia in patients affected by a stroke. Randomized controlled trials (RCTs) comparing acupuncture treatment with non-acupuncture treatment of dysphagia in patients with a stroke were identified from the databases of PubMed, Embase, Cochrane Library and CBM disc (China Biological Medicine Database). Eligible investigations were included and data on the effectiveness of acupuncture were extracted and synthesized by meta-analysis using RevMan 5.1.4. Results were expressed as odds ratio (OR) for dichotomous data; 95 % CIs were also calculated. A total of 72 RCTs (3,208 patients in the treatment group and 2,926 patients in the control group) were identified. Details of randomization and blinding were not reported and information on withdrawals and drop-outs was missing in most of included reports. Meta-analysis showed that the effectiveness of treatment in the group receiving acupuncture was higher than that in the non-acupuncture group (OR = 5.17, 95 % CI: 4.18 to 6.38; $p < 0.00001$). However, the study quality was generally low and of insufficient quality to make recommendations about using acupuncture in the rehabilitation of patients with dysphagia due to stroke. The authors concluded that acupuncture might be beneficial in the rehabilitation of patients with dysphagia caused by stroke, and the evidence justifies future high-quality studies.

Jiang et al (2022) stated that post-stroke dysphagia is one of the most common stroke complications with high morbidity and long course, while acupuncture treatment is easily accepted by patients due to its reliability, feasibility, simple operation, low price, and quick effect. In a systematic review and meta-analysis, these investigators examined the effectiveness of manual acupuncture in patients with post-stroke dysphagia. Databases including Medline, Web of Science, PubMed, Cochrane Library databases, Embase, CNKI (China National Knowledge Infrastructure), WanFang (WanFang Database), and VIP (Chongqing VIP) were searched from inception until August 19, 2022. Data were analyzed using Revman 5.3, Stata 14.0, and TSA 0.9.5.10 Beta software. Evidence quality evaluation was carried out by employing GRADE profiler 3.6. A total of 33 RCTs enrolled 2,680 participants. Meta-analysis revealed that compared to rehabilitation, acupuncture decreased water swallow test (WST) and SSA scores. Meanwhile, in contrast to rehabilitation alone, integration of acupuncture with rehabilitation effectively decreased WST and SSA scores; improved swallowing scores of VFSS, swallowing scores of Fujishima Ichiro, BI, and SWAL-QOL; reduced the aspiration rates and aspiration pneumonia; as well as shortened the duration of empty swallowing and the duration of 5-ml water swallowing. Pooled analysis did not reveal any significant differences in DOSS ($p = 0.15 > 0.05$) between the acupuncture group combined with rehabilitation group and the rehabilitation group alone. After the risk-of-bias assessment, these studies were not of low quality, except in terms of allocation concealment and blindness. Evidence quality

evaluation showed that allocation concealment and blindness led to a down-grade and primary outcomes' evaluation of acupuncture combined with rehabilitation were ranked as moderate-quality evidence while acupuncture alone was ranked as low-quality. The authors concluded that this meta-analysis provided positive pieces of evidence that acupuncture and acupuncture combined with rehabilitation were better than using rehabilitation alone in the treatment of post-stroke dysphagia. Moreover, these researchers stated that multi-center RCTs with a large sample and a rigorous design are needed to examine if acupuncture could replace other stimulative therapies in rehabilitation training. Moreover, acupoint combination, frequency, and the total number of treatments may be important factors that could influence therapeutic effect, which can provide guidance for subsequent similar RCTs.

The authors stated that this review had several drawbacks. First, although most of the included studies used clinical success rate as an outcome indicator, there was no uniform international standard for the definition of success rate. Thus, these researchers did not consider clinical response rate as one of the outcome indicators, which may have caused a loss of evidence. Second, post-stroke dysphagia is one of the most prevalent stroke complications with high morbidity and a long disease course. The shortest course of treatment in the included studies was 10 days, while the longest was 8 weeks; and only 1 study had a 3-month follow-up and reported a long-term effectiveness. Third, although these investigators used Egger's test to examine publication bias in this review, the fact that all the included RCTs were conducted in China potentially contributed to a publication bias.

Biodegradable Stent

Griffiths et al (2012) noted that biodegradable (BD) esophageal stents have been available commercially only since 2008 and previous published research is limited. These researchers reviewed the use of BD stents to treat dysphagia in benign or malignant esophageal strictures. Patients were identified from a prospective interventional radiological database. Biodegradable stents were inserted radiologically under fluoroscopic control. Between July 2008 and February 2011, a total of 25 attempts at placing SX-ELLA BD esophageal stents were made in 17 males and 5 females, with a median age of 69 (range of 54 to 80) years. Two patients required more than 1 BD stent.

Indications were benign strictures (n = 7) and esophageal cancer (n = 17). One attempt was unsuccessful for a technical success rate of 96 % with no immediate complications.

Clinical success rate was 76 %. Median dysphagia score before stent insertion was 3 (range of 2 to 4) compared to 2 (range of 0 to 3) after stent insertion (p = 0.0001). The authors concluded that BD stents provide good dysphagia relief for the life time of the stent. They may help avoid the use of feeding tubes in patients having radical chemoradiotherapy or awaiting esophagectomy. They do not require removal or interfere with radiotherapy planning via imaging. However, the re-intervention rate was high after the stent dissolves.

Krokidis et al (2013) evaluated the clinical results of the use of BD esophageal stents in malignant strictures. A total of 11 patients were included in this prospective analysis in which a woven polydioxanone BD esophageal stent was used. The inclusion criterion was that the patient underwent neoadjuvant treatment or radical radiotherapy after the stent insertion. Primary end points were dysphagia score at discharge, stent patency, and complication rate. Secondary end points were overall survival and surgical outcome of surgery. There was a 100 % procedure technical success rate. Early complications occurred in 3 patients resulting in failure to restore oral nutrition. In the remaining 8 patients, dysphagia was significantly improved at discharge. Mean stent patency rate in this group was 71.5 days. Stent dysfunction occurred in 5 of 8 patients (62.5 %); in 2 of 5 patients this was due to local inflammatory reaction, and in 3 of 5 patients it was due to tumor growth after a mean time of 97.8 days, and a new metallic stent was consequently placed in 4 of 5 patients. One patient was successfully treated with esophagectomy. At the end of follow-up (mean time of 102.1 days), 3 of 8 stents were patent. The overall patient survival rate was 81.8 %. The authors concluded that although short-term dysphagia scores improved, BD stents do not appear to offer a clear beneficial effect in most cases of malignant strictures, particularly due to a local inflammatory reaction that may be induced. Technical improvement of the device and delineation of the patient group that would benefit from its use is necessary if further studies are to be conducted in the future.

Botulinum Toxin Injection

In a pilot study, Terre et al (2008) evaluated the effectiveness of botulinum toxin (BTX-A) injection in the cricopharyngeus muscle in patients with neurological dysphagia caused by alteration in the upper esophageal sphincter (UES) opening and with preserved pharyngeal contraction. A total of 10 patients (7 brain lesions and 3 cervical spinal cord injuries), with a minimum time-lapse of 6 months from neurological lesion received BTX-A injection. Dysfunction of the UES opening and the presence of pharyngeal contraction were diagnosed by videofluoroscopy (VDF) and esophageal manometry (EM). The BTX-A (100 U) injection was guided by endoscopy. Clinical, VDF, and EM follow-ups were carried out at 3 weeks, 3 and 6 months, and at 1 year post-injection. Prior to treatment, 6 patients were fed by naso-gastric tube; VDF showed impairment of the UES opening, residue in pyriform sinuses, and aspiration in all cases. During follow-up, there was a decrease in the number of patients who had aspiration: 3 patients at 1 year. During swallowing, EM showed a mean UES relaxation of 90 % (range of 74.5 to 100 %), residual pressure 3.2 mm Hg (range of 0 to 13 mm Hg) and pharyngeal amplitude 52 mm Hg (range of 25 to 80 mm Hg). At follow-up, a significant improvement in UES relaxation (98 % [89 % to 100 %]) and pharyngeal contraction (97 mm Hg [35 mm Hg to 165 mm Hg]) was observed. At 3 months, 6 patients were eating exclusively by mouth. The authors concluded that 1 single injection of BTX-A in the UES has long-lasting effectiveness in patients with neurological dysphagia caused by alteration in the UES opening and with pharyngeal contraction. Nevertheless, they stated that a randomized control trial should be done to confirm these results and rule out the effect of potential spontaneous improvement of neurological injury.

Electrical Stimulation (Including Neuromuscular Electrostimulation)

Electrical stimulation (ES) has been examined for the treatment of dysphagia. However, there is currently insufficient evidence to support the effectiveness of ES in treating this condition. Park et al (1997) reported a pilot study of oral ES on swallow function in post-stroke patients. They found that oral ES resulted in an improvement in swallow function in 2 of the 4 patients. The authors concluded that these early results are promising, but further research is needed. In a controlled study, Freed et al (2001) compared the effectiveness of transcutaneous ES to thermal-tactile stimulation (TS) in patients with dysphagia caused by stroke. The investigators concluded that transcutaneous ES appears to be a safe and effective treatment for dysphagia due to stroke and results in better swallow function than conventional TS treatment. However, there were no follow-up data in this study. Grill et al (2001) reviewed emerging clinical applications of ES, and concluded that functional ES has great potential for increasing life support as well as for quality of life in chronic ailments, particularly obstructive sleep apnea and dysphagia.

In a non-concurrent cohort study, Blumenfeld et al (2006) assessed the effectiveness of ES in treating persons with dysphagia and aspiration. The charts of 40 consecutive subjects undergoing ES and 40 consecutive persons undergoing traditional dysphagia therapy (TDT) were reviewed. Pre- and post-therapy treatment success was compared utilizing a previously described swallow severity scale. A linear regression analysis was employed to adjust for potential confounding variables. The swallow severity scale improved from 0.50 to 1.48 in the TDT group ($p < 0.05$) and from 0.28 to 3.23 in the ES group ($p < 0.001$). After adjusting for potential confounding factors, persons receiving ES did significantly better in regard to improvement in their swallowing function than persons receiving TDT ($p = 0.003$). The authors concluded that the findings of this non-concurrent cohort study suggested that dysphagia therapy with transcutaneous ES is superior to traditional dysphagia therapy alone in individuals in a long-term acute care facility. They also stated that confirmation of these findings with a prospective, placebo-controlled, randomized clinical trial is needed before a definitive determination regarding the effectiveness of ES dysphagia therapy can be made.

Kiger et al (2006) compared the outcomes using transcutaneous neuromuscular ES (VitalStim therapy) to outcomes using traditional swallowing therapy for deglutition disorders. A total of 22 patients had an initial and a follow-up video-fluoroscopic swallowing study or fiberoptic endoscopic evaluation of swallowing and were divided into an experimental group that received VitalStim treatments and a control group that received traditional swallowing therapy. Outcomes were analyzed for changes in oral and pharyngeal phase dysphagia severity, dietary consistency restrictions, and progression from non-oral to oral intake. Results of chi-square analysis showed no statistically significant difference in outcomes between the experimental and control groups.

Shaw and colleagues (2007) carried out a retrospective analysis of 18 patients with

dysphagia who received VitalStim therapy. All subjects underwent pre-therapy evaluation by speech-language pathologists, including modified barium swallow and/or functional endoscopic evaluation of swallowing and clinical evaluation of swallowing that included assessment of laryngeal elevation, diet tolerance, and swallowing delay, and were then assigned an overall dysphagia severity score. After therapy, all patients underwent the same assessments. Twelve of the 18 subjects also underwent a functional swallowing telephone survey months (range of 1 to 21 months) after their therapy to evaluate if the improvement was worthwhile and sustained. Eleven of the 18 patients (61 %) demonstrated some improvement in their swallowing; 6 of the 18 patients (33 %) were improved enough to no longer require a feeding tube. However, of the 5 patients categorized as having "severe dysphagia" before therapy, only 2 showed any improvement, and these patients still required a feeding tube for adequate nutrition. Telephone surveys did confirm that those who improved with their therapy seemed to maintain their progress and that most patients were satisfied with their therapy. The authors concluded that VitalStim therapy seems to help those with mild-to-moderate dysphagia. However, the patients with the most severe dysphagia did not gain independence from their feeding tubes.

Miller et al (2014) performed a systematic literature search in the Cochrane Central Register of Controlled Trials, the Cochrane Database of Systematic Reviews, the DAHTA database, the Health Technology Assessment Database and MEDLINE or PubMed considered studies on the use of neuromuscular electrostimulation (NMES) in otorhinolaryngology that have been published in German or English. The search identified 180 studies. These were evaluated and relevant studies were included in the further evaluation. The authors concluded that in the fields of otorhinolaryngology and phoniatry/pediatric audiology, clinical studies investigating the effects of NMES on facial and laryngeal paresis, as well as dysphonia and dysphagia have been carried out. The evidence collected to date is encouraging; particularly for the treatment of certain forms of dysphagia and laryngeal paresis.

Frost and associates (2018) stated that previous research has suggested that treatment using NMES when used in conjunction with conventional therapy is effective. These investigators described the recent literature and a small prospective case series performed in the United Kingdom (UK). This study contributed to support National Institute of Clinical Excellence (NICE) guidance for clinicians who wish to include NMES in a rehabilitation program for dysphagic patients, specifically with reference to safety and the impact on swallowing function of this intervention. In 2014, the UK NICE issued guidelines enabling UK therapists to trial the use of NMES, but the guidelines also sought additional evidence on the impact on swallowing function of NMES and the incidence of side effects. This small prospective case series investigated both of these aspects with a group of patients with dysphagia of neurological origin who had not achieved adequate swallowing function with traditional therapy alone. This study recruited 10 adult patients with dysphagia of neurological origin. All had previously received traditional swallowing therapy for at least 6 months but only achieved a Functional Oral Intake Scale (FOIS) of 4 or less (a scale for amounts and types of oral intake). The total study period was 10 weeks for each subject comprising 5 weeks of traditional therapy delivered 3 times a week followed by 5 weeks of NMES concurrent with traditional therapy (NMES + traditional therapy) delivered 3 times a week using the VitalStim stimulator (VitalStim Therapy, UK). In addition, the Quality of Life in Swallowing and Eating Assessment Tool 10 (QOL scales) were determined to allow comparison between studies. A total of 9 subjects achieved an improvement in swallowing function after NMES + traditional therapy was measured using the FOIS, giving a statistically significant improvement ($p < 0.001$) when NMES + traditional therapy was compared with traditional therapy. In addition, there was a low incidence of AEs with only minor AEs occurring in 1.3 % of electrode pair placements. The authors concluded that the current preliminary study suggested that NMES + traditional therapy had a good safety record and may improve oral intake and patient reported swallowing outcomes for subjects with long-standing dysphagia of neurological cause, which has not responded to traditional therapy. They stated that provided care is taken with skin preparation and electrode placement, the risk of AEs from the treatment was minimal. Significantly, in 30 % of the subjects, an improvement in voice quality was found. These findings need to be validated by well-designed studies.

In a randomized, single-blind, placebo-controlled trial, Park and colleagues (2018) examined the effect of effortful swallowing combined with NMES as a novel treatment approach in dysphagic patients with Parkinson's disease. Participants were randomly allocated to an experimental group ($n = 9$) or a placebo group ($n = 9$). The experimental group simultaneously received NMES with effortful swallowing, while the placebo group

received sham NMES with effortful swallowing. All participants received the treatment for 30 mins/day at 5 sessions per week for 4 weeks. Both groups also received the same conventional dysphagia therapy. The experimental group showed significant differences in horizontal movement ($p = 0.038$) and vertical movement ($p = 0.042$) compared to the placebo group, but showed no significant differences in the oral ($p = 0.648$) or pharyngeal phase ($p = 0.329$) of the video-fluoroscopic dysphagia scale compared to the placebo group, except for the penetration-aspiration scale ($p = 0.039$). The authors concluded that the findings of this study demonstrated that NMES applied to the infra-hyoid region combined with effortful swallowing effectively increased hyoid bone movement and reduced aspiration in dysphagic patients with Parkinson's disease. These findings need to be validated by studies with larger sample size.

Per-Oral Endoscopic Myotomy (POEM)

The American college of Gastroenterology (ACG)'s clinical guideline on "Diagnosis and management of achalasia" (Vaezi et al, 2013) identified POEM as experimental. It stated that "Randomized prospective comparison trials with standard laparoscopic myotomy and/or PD [pneumatic dilation] are needed and POEM should only be performed in the context of clinical trials with the understanding that other effective well-studied alternatives are available".

Von Renteln et al (2013) stated that pilot studies have indicated that per-oral endoscopic myotomy (POEM) might be a safe and effective treatment for achalasia. These investigators performed a prospective, international, multi-center study to determine the outcomes of 70 patients who underwent POEM at 5 centers in Europe and North America. Three months after POEM, 97 % of patients were in symptom remission (95 % CI: 89 % to 99 %); symptom scores were reduced from 7 to 1 ($p < 0.001$) and lower esophageal sphincter (LES) pressures were reduced from 28 to 9 mm Hg ($p < 0.001$). The percentage of patients in symptom remission at 6 and 12 months was 89 % and 82 %, respectively. The authors concluded that POEM was found to be an effective treatment for achalasia after a mean follow-up period of 10 months. The main drawbacks of this study were the lack of a control group and the short-term follow-up.

In a prospective trial, Verlaan et al (2013) evaluated the effect of POEM on esophagogastric function. Patients were greater than 17 years of age with achalasia and an Eckardt score of greater than or equal to 3. Before and 3 months after POEM, 10 consecutive patients underwent esophageal manometry, timed barium esophagograms, and EndoFLIP as well as an esophago-gastro-duodenoscopy (EGD). Main outcome measures were Eckardt symptom score, LES resting pressure, centimeters of barium stasis, esophagogastric junction (EGJ) distensibility, and reflux esophagitis. Compared with scores before POEM, patient symptom scores were significantly reduced (1, interquartile range [IQR 0 to 1] versus 8 [IQR 4 to 8]; $p = 0.005$). Lower esophageal sphincter pressure decreased significantly (6.0 mm Hg [IQR 2.6 to 7.4] versus 19.0 mm Hg [IQR 13.0 to 28.0]; $p = 0.008$). Esophageal emptying increased significantly, and a 5-min barium column measured 2.3 cm (IQR 0 to 3.2 cm) versus 10.1 cm (IQR 5.7 to 10.8 cm; $p = 0.005$). Esophagogastric junction distensibility increased significantly (6.7 mm(2)/mm Hg [IQR 3.8 to 16.6] versus 1.0 mm(2)/mm Hg [IQR 0.4 to 2.3]; $p = 0.02$) at 50 ml. In 6 of 10 patients, reflux esophagitis was seen. Of these patients, 3 reported reflux symptoms. The authors concluded that POEM improves esophagogastric function and suggested favorable long-term results based on Eckardt score, esophageal manometry, esophageal emptying, and EGJ distensibility. Moreover, they stated that long-term follow-up of larger series will determine whether the high rate of reflux esophagitis affects the clinical application of POEM. The main drawbacks of this study were small number of patients, and short-term follow-up.

Onimaru et al (2013) evaluated the safety and effectiveness of POEM for surgical myotomy failure as a rescue second-line treatment, and discussed the treatment options adapted in achalasia recurrence. A total of 315 consecutive achalasia patients received POEM from September 2008 to December 2012 in the authors' hospital. A total of 11 (3.5 %) patients, who had persistent or recurrent achalasia and had received surgical myotomy as a first-line treatment from other hospitals, were included in this study. Patient background, barium swallow studies, EGD, manometry, and symptom scores were prospectively evaluated. In principle, all patients in whom surgical myotomy failed received pneumatic balloon dilatation (PBD) as the first line "rescue" treatment, and only if PBD failed were patients considered for rescue POEM. The PBD alone was effective in 1 patient, and in the remaining 10 patients, rescue POEM was performed successfully without complications. Three months after rescue POEM, significant reduction in LES resting

pressures (22.1 ± 6.6 mm Hg versus 10.9 ± 4.5 mm Hg, $p < 0.01$) and Eckardt symptom scores (6.5 ± 1.3 versus 1.1 ± 1.3 , $p < 0.001$) were observed. The authors concluded that short-term results of POEM for failed surgical myotomy were excellent; long-term results are awaited.

Yang and Wagh (2013) stated that achalasia is a motility disorder of the esophagus, characterized by a peristalsis of the esophageal body and incomplete relaxation of the LES. Treatment of achalasia is currently aimed at decreasing the resting pressure in the LES. Per-oral endoscopic myotomy is an emerging novel endoscopic procedure for the treatment of achalasia with initial data suggesting an acceptable safety profile, excellent short-term symptom resolution, low incidence of post-procedural gastro-esophageal reflux (GER), and improvement in manometric outcomes. The authors concluded that further prospective randomized trials are needed to evaluate the long-term effectiveness of this promising technique compared to other treatment modalities for achalasia.

Friedel et al (2013) stated that the volume of POEMs performed worldwide has grown exponentially. In fact, surgeons who have performed Heller myotomy have embraced POEM as the preferred intervention for achalasia. However, the authors stated that the niche of POEM remains to be defined and long-term results are awaited.

Pescarus et al (2014) stated that POEM is a new minimally invasive endoscopic treatment for achalasia. Since the first modern human cases were published in 2008, around 2,000 cases have been performed worldwide. This technique requires advanced endoscopic skills and a learning curve of at least 20 cases. Per-oral endoscopic myotomy is highly successful with over 90 % improvement in dysphagia while offering patients the advantage of a low impact endoscopic access. The main long-term complication is GER with an estimated incidence of 35 %, similar to the incidence of GER post-laparoscopic Heller with fundoplication. The authors concluded that although POEM represents a paradigm shift in the treatment of achalasia, more long-term data are needed to further define its role in the treatment algorithm of this rare disease.

Bredenoord et al (2014) noted that treatment of achalasia is complicated by symptom recurrence and a significant risk for severe complications. Endoscopic myotomy was developed in the search for a highly effective treatment with lower risks. Since its introduction in 2010, several centers have adopted the technique and published excellent short-term results of open label series. These researchers stated that randomized trials with long-term end-point comparing POEM with the established treatments such as balloon dilation and surgical myotomy are now warranted, before POEM can be regarded as the routine clinical care for achalasia patients.

Furthermore, an UpToDate review on "Overview of the treatment of achalasia" (Spechler, 2014) states that "Long-term data from randomized trials are necessary to compare POEM with laparoscopic surgical myotomy and pneumatic dilation before POEM can be recommended".

Schlottmann et al (2018) compared the outcome of per oral endoscopic myotomy (POEM) and laparoscopic Heller myotomy (LHM) for the treatment of esophageal achalasia. These researchers performed a systematic Medline literature search of articles on LHM and POEM for the treatment of achalasia. The main outcomes measured were improvement of dysphagia and post-treatment gastro-esophageal reflux disease (GERD). Linear regression was used to model the effect of each procedure on the different outcomes. A total of 53 studies reported data on LHM (5,834 patients), and 21 articles examined POEM (1,958 patients). Mean follow-up was significantly longer for studies of LHM (41.5 versus 16.2 months, $p < 0.0001$). Predicted probabilities for improvement in dysphagia at 12 months were 93.5 % for POEM and 91.0 % for LHM ($p = 0.01$), and at 24 months were 92.7 % for POEM and 90.0 % for LHM ($p = 0.01$). Patients undergoing POEM were more likely to develop GERD symptoms (odds ratio [OR] 1.69, 95 % confidence interval [CI]: 1.33 to 2.14, $p < 0.0001$), GERD evidenced by erosive esophagitis (OR 9.31, 95 % CI: 4.71 to 18.85, $p < 0.0001$), and GERD evidenced by pH monitoring (OR 4.30, 95 % CI: 2.96 to 6.27, $p < 0.0001$). On average, length of hospital stay was 1.03 days longer after POEM ($p = 0.04$). The authors concluded that short-term results showed that POEM is more effective than LHM in relieving dysphagia, but it is associated with a very high incidence of pathologic reflux. The clinical sequelae of the increase in pathologic reflux are currently unclear, but it is possible that longer-term outcomes associated with POEM may demonstrate GERD complications such as stricture and/or Barrett esophagus. These researchers stated that longer follow-up studies and randomized trials comparing POEM to LHM are needed to establish the role of this new technique in the treatment algorithm of

achalasia.

The authors stated that this study had several drawbacks. Due to the rarity of achalasia, few controlled trials were available for analysis and most of the studies reviewed by these investigators were retrospective or cohort studies, subject to selection bias and reporting bias. In addition, POEM studies have significant shorter follow-up as compared with LHM studies, which may influence the results. As noted above, POEM is also a newer technology, and the extent to which outcomes change as operators exit their learning curves is unclear. Heterogeneity in reporting outcomes among studies may have also affected the results, and limited the categorization of outcomes into simple dichotomous variables that could be easily quantified. LHM studies in general evaluated their primary outcomes by the improvement in the swallowing status; POEM studies, on the other hand, most commonly assessed outcomes using the Eckardt score, a more precise evaluation as it considers not only the improvement in dysphagia, but also the effect of the procedure on regurgitation, retro-sternal pain and weight loss.

In a systematic review and meta-analysis, Awaiz et al (2017) compared the safety and effectiveness of LHM and POEM for the treatment of achalasia. These investigators carried out a search of PubMed, Cochrane database, Medline, Embase, Science Citation Index, and current contents for English-language articles comparing LHM and POEM between 2007 and 2016. Variables analyzed included prior endoscopic treatment, prior medical treatment, prior Heller myotomy, operative time, overall complications rate, post-operative GERD, length of hospital stay, post-operative pain score, and long-term GERD. A total of 7 trials consisting of 483 (LHM = 250, POEM = 233) patients were analyzed. Pre-operative variables (e.g., prior endoscopic treatment [OR, 1.32; 95 % CI: 0.23 to 4.61; $p = 0.96$], prior medical treatment [weighted mean difference (WMD), 1.22; 95 % CI: 0.52 to 2.88; $p = 0.65$], and prior Heller myotomy (WMD, 0.47; 95 % CI: 0.13 to 1.67; $p = 0.25$) were comparable. Operative time was 26.28 minutes, non-significantly longer for LHM (WMD, 26.28; 95 % CI: -11.20 to 63.70; $p = 0.17$). There was a comparable overall complication rate (OR, 1.25; 95 % CI: 0.56 to 2.77; $p = 0.59$), post-operative GERD rate (OR, 1.27; 95 % CI: 0.70 to 2.30; $p = 0.44$), length of hospital stay (WMD, 0.30; 95 % CI: -0.24 to 0.85; $p = 0.28$), post-operative pain score (WMD, -0.26; 95 % CI: -1.58 to 1.06; $p = 0.70$), and long-term GERD (WMD, 1.06; 95 % CI: 0.27 to 4.1; $p = 0.08$) for both procedures. There was a significantly higher short-term clinical treatment failure rate for LHM (OR, 9.82; 95 % CI: 2.06 to 46.80; $p < 0.01$). The authors concluded that POEM compared favorably to LHM for achalasia treatment in short-term peri-operative outcomes. However, there was a significantly higher clinical treatment failure rate for LHM on short-term post-operative follow-up. They stated that currently long-term post-operative follow-up data for POEM beyond 1 year are unavailable and eagerly awaited. They stated that presently, long-term post-operative follow-up data for POEM beyond 1 year are unavailable in all the comparative trials and therefore meaningful comparison with LHM is impossible. In the future, several issues need to be addressed to determine the clinical outcomes, safety, and effectiveness of these 2 methods for achalasia treatment. These include the following:

These researchers stated that until all the above-mentioned issues are rigorously addressed in a well-designed randomized controlled trial comparing POEM with LHM, the routine use of POEM should proceed with caution.

Crespin et al (2017) performed a systematic review of the literature to evaluate the safety and effectiveness of POEM for the treatment of achalasia. The systematic review was conducted following the PRISMA guidelines. Evidence-Based Medicine Reviews, Cochrane Central Register of Controlled Trials, Ovid MEDLINE (R) including in-process and non-indexed citations were searched for POEM studies using the keywords: esophageal achalasia, POEM, endoscopy, natural orifice surgery, laparoscopic Heller myotomy (LHM), and related terms. Eckardt score, lower esophageal sphincter (LES) pressure, and reported complications were the main outcomes. Two authors reviewed the search result independently. A third reviewer resolved all disagreements. Data abstraction was pilot-tested and approved by all authors. Data were examined for clinical, methodological, and statistical heterogeneity with the aim of determining whether evidence synthesis using meta-analysis was possible. The search strategy retrieved 2,894 citations. After removing duplicates and applying the exclusion criteria, 54 studies were selected for full-text review of which a total of 19 studies were considered eligible for further analysis. There were 10 retrospective and 9 prospective studies, including 1,299 POEM procedures. No randomized control trial (RCT) was identified. Overall, the pre- and post-POEM Eckardt scores and LES pressure were significantly different. The most

frequently reported complications were mucosal perforation, subcutaneous emphysema, pneumoperitoneum, pneumothorax, pneumomediastinum, pleural effusion, and pneumonia. The median follow-up was 13 months (range of 3 to 24). The authors concluded that POEM is a safe and effective alternative for the treatment of achalasia. However, only short-term follow-up data compared with LHM are available. These investigators stated that RCTs and long-term follow-up studies are needed to establish the safety and effectiveness of POEM in the management of patients with achalasia.

Hernandez Mondragon et al (2017) noted that POEM is an excellent endoscopic treatment for achalasia. Clinical and manometric parameters are used for evaluation and follow-up. However, clinical success does not guarantee high QOL scores, generating doubts about their direct relationship. These researchers evaluated QOL scores before and after POEM at medium- and long-term, to evaluate differences between achalasia subtypes and find which factors related to low QOL scores. Achalasia-confirmed patients undergoing POEM between February 2012 and November 2016, and completing at least 1 year of follow-up, were included. Assessment before and at 1, 6, 12, 24, 36 and 48 months after POEM employed manometry, barium series, Eckardt score, and the AE-18 health-related QOL scale. Demographic, clinical, and procedure characteristics were documented, with comparisons between subgroups. Multiple logistic regression analysis was done; 65 of 88 patients were included (38 women and 27 men; median age of 47 years, interquartile range [IQR] 20 to 81), and 50 (76.9 %) completed 4 years of follow-up. Eckardt score improved (median, pre-procedure 10 versus post-procedure 2; $p = 0.002$) and this persisted. There was initial improvement in median integrated relaxation pressure (IRP) (29.4 mmHg [16 to 55] versus 10.3 mmHg [3 to 18]; $p = 0.000$) and median QOL scores (40 versus 68 at 1 month; $p = 0.002$); however IRP increased and QOL scores decreased. Men with confirmed type III achalasia had low QOL scores. The authors concluded that all patients had significant clinical improvement after POEM, with medium- to long-term persistence. These investigators noted that although quality of life and IRP initially improved, they deteriorated in the long-term; male sex and type III achalasia appeared to be associated with low QOL scores.

Liu et al (2017) stated that with the development of endoscopic interventions and inspired by the success of POEM for the treatment of achalasia, these researchers investigated an old method of direct POEM without a submucosal tunnel for the treatment of achalasia, which these investigators called open POEM (O-POEM). In this study, clinical success was achieved in the patient after O-POEM. A reduction of LES pressure, Eckardt score, and a timed barium esophagogram were observed during follow-up. There were no severe complications and no recurrences during 2 months of follow-up. The authors concluded that O-POEM is a feasible and effective endoscopic treatment modality for achalasia. However, long-term outcomes of O-POEM requires further follow-up.

Banks and Sweis (2017) stated that achalasia is a rare esophageal motility disorder predominantly causing dysphagia and regurgitation of food and fluids. Diagnosis is made typically after a combination of tests including endoscopy, barium swallow and esophageal manometry. The advent of high-resolution manometry has led to the Chicago Classification that divided achalasia into 3 types. This improved the understanding of presentation, prognosis and might also help tailor therapy. Botulinum toxin has been shown to have good, but short-term efficacy. The predominant treatments include pneumatic balloon dilatation and laparoscopic Heller's myotomy, both of which have similar and durable outcomes, although the success of both reduces with time; POEM has been shown to be as effective, safe and durable as earlier treatments for achalasia; however, RCTs are lacking. The authors noted that indications for POEM are expanding to other hyper-contraction motility disorders of the esophagus.

Nabi et al (2017) stated that POEM has emerged as an efficacious treatment modality for the management of achalasia cardia (AC) and non-achalasia spastic esophageal motility disorders. Initial results are encouraging. These investigators analyzed the safety and effectiveness of POEM in a large cohort of patients with AC. The data from patients who underwent POEM (from January 2013 to June 2016) was prospectively collected and analyzed. Clinical success was defined as Eckardt score less than or equal to 3 after POEM procedure. Objective parameters including high-resolution manometry (HRM) and timed barium swallow (TBS) were analyzed and compared before and after the procedure. Gastro-esophageal reflux was analyzed using 24-hour pH impedance study and esophago-gastro-duodenoscopy. A total of 408 patients (mean age of 40 years, range of 4 to 77 years) underwent POEM during the specified period. POEM could be successfully completed in 396 (97 %) patients. Clinical success rates at 1, 2 and 3 years were 94 %, 91 % and 90 %, respectively. Mean Eckardt score was 7.07 ± 1.6 prior to POEM and 1.27 ± 1.06 after POEM

($p = 0.001$) at 1 year. Significant improvement in esophageal emptying on TBE (greater than 50 %) was documented in 93.8 % patients who completed 1-year follow up. Pre-procedure and post-procedure mean lower esophageal sphincter pressure was 45 ± 16.5 mmHg and 15.6 ± 6.1 mmHg, respectively ($p = 0.001$). Technical and clinical success were comparable in naïve versus prior treated cases (97.3 % versus 96.8 %, $p = 0.795$) (95.7 % versus 92.6 %, $p = 0.275$). GERD was documented in 28.3 % patients with 24-hour pH-impedance study and erosive esophagitis was seen in 18.5 % of patients who underwent POEM. The authors concluded that POEM is safe, effective and durable for treatment of AC. The incidence of GERD did not appear to be higher than with LHM. However, randomized comparisons are needed; whether POEM should be offered as a 1st-line treatment to all patients with AC is a matter of debate. Moreover, they stated that long-term follow-up studies and randomized comparison with established modalities such as PBD and LHM will provide conclusive information in that regard.

Schlottmann and Patti (2018) noted that esophageal achalasia is a rare disorder characterized by a failure of the LES to relax during swallowing, combined with aperistalsis of the esophageal body. Treatment is not curative, but aims to eliminate the outflow resistance caused by the non-relaxing LES. Current evidence suggests that both laparoscopic Heller myotomy and POEM are very effective in the relief of symptoms in patients with achalasia. Specifically, for type III achalasia, POEM may achieve higher success rates. However, POEM is associated to a very high incidence of pathologic reflux, with the risk of exchanging one disease-achalasia with another gastro-esophageal reflux.

Kroch and Grimm (2018) stated that POEM was introduced in 2008 for the treatment of esophageal achalasia. It is performed endoscopically, which allows transection of the muscular fibers of the distal esophagus and of the LES. The procedure is therefore similar to a laparoscopic Heller myotomy without a fundoplication. Short-term studies have shown that POEM is very effective in relieving dysphagia and regurgitation, but concerns have been raised about the incidence of post-POEM gastro-esophageal reflux. The authors concluded that prospective and randomized trials are needed to determine the role of this new procedure in the treatment algorithm of esophageal achalasia.

Furthermore, an UpToDate review on "Overview of the treatment of achalasia" (Spechler, 2018) states that "Per-oral endoscopic myotomy (POEM) is a promising new endoscopic technique for performing myotomy of the LES. Good results for POEM have been reported in patients with achalasia conditions that often do not respond well to conventional therapies such as type III (spastic) achalasia and "end stage" achalasia (markedly dilated, sigmoid esophagus), and in patients who have failed prior endoscopic and surgical achalasia treatments. However, long-term data on the efficacy of POEM are very limited, and there are no randomized trials that compare POEM with the conventional achalasia treatments of laparoscopic surgical myotomy and pneumatic dilation. Consequently, the role of POEM in the treatment of achalasia has not been established".

Repici and colleagues (2018) stated that POEM represents a less invasive alternative to conventional laparoscopic Heller's myotomy (LHM) for patients with achalasia. However, it cannot be excluded that the lack of fundoplication after POEM may result in a higher incidence of reflux disease, as compared with LHM. These researchers conducted a systematic review of prospective studies reporting the incidence of reflux disease developed after POEM and LHM. A literature search with electronic databases was performed (up to February 2017) to identify full articles on the incidence of gastro-esophageal reflux symptoms and endoscopic monitoring and pH monitoring findings after POEM and LHM (with fundoplication). Proportions and rates were pooled by means of random or fixed-effects models, according to the level of heterogeneity between studies. After the selection criteria was applied, 17 and 28 studies, including 1,542 and 2,581 participants who underwent POEM and LHM, respectively, were included. The pooled rate of post-procedural symptoms was 19.0 % (95 % confidence interval [CI]: 15.7 % to 22.8 %) after POEM and 8.8 % (95 % CI: 5.3 % to 14.1 %) after LHM, respectively. The pooled rate estimate of abnormal acid exposure at pH monitoring was 39.0 % (95 % CI: 24.5 % to 55.8%) after POEM and 16.8 % (95 % CI: 10.2 % to 26.4 %) after LHM, respectively. The rate of esophagitis after POEM was 29.4 % (95 % CI: 18.5 % to 43.3 %) after POEM and 7.6 % (95 % CI: 4.1 % to 13.7 %) after LHM. At meta-regression, heterogeneity was explained partly by the POEM approach and study population. The authors concluded that the incidence of reflux disease appeared to be significantly more frequent after POEM than after LHM with fundoplication. Monitoring pH and ensuring appropriate treatment after POEM should be considered in order to prevent long-term reflux-related adverse events.

Cho and Kim (2018) noted that POEM is an endoluminal procedure that involves dissection

of esophageal muscle fibers followed by submucosal tunneling. Inoue first attempted to use POEM for the treatment of achalasia in humans. Expanded indications of POEM include classic indications such as type I, type II, type III achalasia, failed prior treatments, including botulinum toxin injection, endoscopic balloon dilation, laparoscopic Heller myotomy, and hypertensive motor disorders such as diffuse esophageal spasm, jackhammer esophagus. Contraindications include prior radiation therapy to the esophagus and prior extensive esophageal mucosal resection/ablation involving the POEM field. Most of the complications are minor and self-limited and can be managed conservatively. As POEM emerged as the main treatment for achalasia, various adaptations to tunnel endoscopic surgery have been attempted. Tunnel endoscopic surgery includes POEM, peroral endoscopic tumor resection, gastric peroral endoscopic pyloromyotomy. The authors concluded that POEM is a safe and effective option for treating type I, type II, type III achalasia, and even for specific cases such as achalasia with failed prior treatments, botulinum toxin injection, endoscopic balloon dilation, LHM and hypertensive motor disorders. However, they stated that large-scale studies and long-term outcomes are needed; OEM has expanded the scope of adaptation to tunnel endoscopic surgeries such as POET and G-POEM, which also require larger scale studies with long-term outcomes.

van Lennep et al (2018) stated that achalasia is a rare esophageal motility disorder. Much of the literature is based on the adult population. In adults, guidance of therapeutic approach by manometric findings has led to improvement in patient outcome. Promising results have been achieved with novel therapies such as POEM. In this review, we provide an overview of the novel diagnostic and therapeutic tools for achalasia management and in what way they will relate to the future management of pediatric achalasia. These investigators performed a PubMed and Embase search of English literature on achalasia using the keywords "children", "achalasia", "pneumatic dilation", "myotomy" and "POEM". Cohort studies of less than 10 cases and studies describing patients greater than or equal to 20 years were excluded. Data regarding patient characteristics, treatment outcome and adverse events (AEs) were extracted and presented descriptively, or pooled when possible. The authors concluded that available data reported that pneumatic dilation and laparoscopic Heller's myotomy are effective in children, with certain studies suggesting lower success rates in pneumatic dilation; POEM is increasingly used in the pediatric setting with promising short-term results; moreover, gastro-esophageal reflux disease (GERD) may occur post-achalasia intervention due to disruption of the LES and therefore requires diligent follow-up, especially in children treated with POEM.

Yeung et al (2018) noted that experience of POEM for treatment of achalasia in pediatric population is limited with varying techniques in different centers. The accurate extent of submucosal tunneling into the gastric cardia and the adequacy of myotomy are the important determining factors to success of POEM. A majority of studies in pediatric population have described using submucosal dye injection for assessing the adequacy of myotomy, however, this is a rather crude and inaccurate method. These investigators described the first case of pediatric achalasia managed with POEM incorporated with novel combined techniques using EndoFLIP (Endoluminal Functional Lumen Imaging Probe) and double endoscope. Esophagogastric junction (EGJ) was identified with a gastroscope. Before POEM, EndoFLIP showed EGJ distensibility index of 1.7 mm²/mmHg. Submucosal tunnel was created from the mucosal entry site at mid-esophagus down and approximately 3 cm beyond the EGJ. Anterior myotomy cutting the circular muscle layer while preserving the longitudinal muscle was performed for 8 cm. Double-endoscope technique was used to confirm the adequacy of myotomy by inserting a smaller endoscope through nostril into stomach and observing the trans-illumination of the first endoscope at the end of submucosal tunnel. After POEM, repeat EndoFLIP measurements revealed increased distensibility index to 6.0 mm²/mmHg. Endoscopic examination at the end of the procedure showed widely patent EGJ. Eckardt symptoms score improved from 9 to 0. At 7 month after POEM, esophagoscopy showed widely open EGJ with no esophagitis, and high-resolution esophageal manometry revealed normalized lower esophageal sphincter pressure and resting tone. The authors concluded that they introduced the intra-operative use of EndoFLIP system that allowed real-time assessment of EGJ distensibility and immediate treatment effect evaluation. Incorporation of double-endoscope POEM was also first described in these pediatric patient for ensuring complete gastric myotomy. They stated that from their experiences, POEM for achalasia in pediatric population appeared to have encouraging results similar to adult patients.

Per-Oral Endoscopic Myotomy (POEM) for the Treatment Type III Achalasia

In an update of the clinical practice from the American Gastroenterological Association

(AGA) on "The use of per-oral endoscopic myotomy in achalasia", Kahrilas and colleagues (2017) described a place for per-oral endoscopic myotomy (POEM) among the currently available robust treatments for achalasia. The recommendations outlined were based on expert opinion and on relevant publications from PubMed and Embase. The Clinical Practice Updates Committee of the AGA proposed the following recommendations: First, in determining the need for achalasia therapy, patient-specific parameters (Chicago Classification subtype, comorbidities, early vs late disease, primary or secondary causes) should be considered along with published efficacy data; second, given the complexity of this procedure, POEM should be performed by experienced physicians in high-volume centers because an estimated 20 to 40 procedures are needed to achieve competence; third, if the expertise is available, POEM should be considered as primary therapy for type III achalasia; fourth, if the expertise is available, POEM should be considered as treatment option comparable with laparoscopic Heller myotomy for any of the achalasia syndromes; and fifth, post-POEM patients should be considered high-risk to develop reflux esophagitis and advised of the management considerations (potential indefinite proton pump inhibitor therapy and/or surveillance endoscopy) of this before undergoing the procedure.

In a review on "Current treatment of achalasia", von Rahden (2019) stated that advantages of POEM are the possibility to perform a long-myotomy (of the entire length of the esophagus if necessary) and the relatively free choice of the localization of the myotomy (anterior/posterior POEM). The disadvantage is the increased post-operative GERD following POEM; however this sequel is managed with proton pump inhibitor (PPI) in most cases, or a laparoscopic fundoplication, if necessary. Preliminary results of 2 prospective, randomized trials showed the superiority of POEM over endoscopic PD, as well as the non-inferiority to LHM, but increased post-operative GERD. The author uses a tailored approach, with preference of POEM for achalasia type III and type II with chest pain and LHM + Dor procedure for sigmoid achalasia and other associated morphological changes.

Zaninotto and associates (2019) evaluated the efficacy, morbidity and side-effects of innovative management strategies for achalasia that include high-resolution manometry (HRM), pneumatic dilatation, laparoscopic Heller's myotomy (LHM), injection of BTX into the LES and POEM. High-resolution manometry has enabled identification of achalasia subtypes that have important prognostic implications. Pneumatic dilatation is a commonly-used and cost-effective method of treating achalasia but has shown poor longevity of symptom relief compared with other modalities and carries a risk of esophageal perforation; LHM is often the preferred, most effective treatment modality, however new studies may show that outcomes are equivalent or even inferior to POEM; BTX injection of the LES has a waning and short duration of efficacy and is used primarily for patients unsuitable for more definitive invasive procedures; POEM is considered the most effective treatment for type III achalasia but carries a high risk of iatrogenic GERD that might predispose to the development of Barrett's esophagus. The authors concluded that HRM and POEM are 2 major innovations in the management of achalasia developed over the past 10 years. There are now 3 major management options for patients with achalasia, namely pneumatic dilatation, LHM and POEM.

Furthermore, an UpToDate review on "Overview of the treatment of achalasia" (Spechler, 2019) states that "Most published data on POEM have come from a relatively small number of highly specialized centers. The largest single-center experience, involving 500 patients with achalasia, described no deaths, but there were adverse procedure-related events (including pneumothorax, bleeding, mucosal perforations and pleural effusions) in 3 % of cases. Two months after POEM, there were significant reductions in symptom scores and LES pressures. However, GERD symptoms at 2 months and at 3 years were noted in 17 and 21 % of patients, respectively. Similar results have been described in other centers. Surprisingly good results for POEM also have been reported in patients with achalasia conditions that often do not respond well to conventional therapies such as type III (spastic) achalasia and "end stage" achalasia (markedly dilated, sigmoid esophagus), and in patients who have failed prior endoscopic and surgical achalasia treatments. At this time, long-term data on the efficacy of POEM are limited, and there are no randomized trials that compare POEM with the conventional achalasia treatments of laparoscopic surgical myotomy and pneumatic dilation. It is not clear that the excellent results reported by highly specialized centers can be duplicated by less experienced endoscopists, and there is controversy regarding the frequency and importance of the GERD that occurs after POEM. For all these reasons, the role of POEM in the treatment of achalasia remains disputed, although there is growing consensus that POEM is the procedure of choice for the treatment of type III achalasia".

Werner et al (2019) noted that pneumatic dilation and laparoscopic Heller's myotomy (LHM) are established treatments for idiopathic achalasia. Per-oral endoscopic myotomy (POEM) is a less invasive therapy with promising early study results. In a randomized, multi-center study, these researchers compared POEM with LHM plus Dor's fundoplication in patients with symptomatic achalasia. The primary endpoint was clinical success, defined as an Eckardt symptom score of 3 or less (range of 0 to 12, with higher scores indicating more severe symptoms of achalasia) without the use of additional treatments, at the 2-year follow-up; a non-inferiority margin of -12.5 percentage points was used in the primary analysis. Secondary endpoints included adverse events (AEs), esophageal function, Gastrointestinal Quality of Life Index score (range of 0 to 144, with higher scores indicating better function), and gastro-esophageal reflux (GER). A total of 221 patients were randomly assigned to undergo either POEM (112 patients) or LHM plus Dor's fundoplication (109 patients). Clinical success at the 2-year follow-up was observed in 83.0 % of patients in the POEM group and 81.7 % of patients in the LHM group (difference, 1.4 percentage points; 95 % confidence interval [CI]: -8.7 to 11.4; $p = 0.007$ for non-inferiority). Serious AEs occurred in 2.7 % of patients in the POEM group and 7.3 % of patients in the LHM group. Improvement in esophageal function from baseline to 24 months, as assessed by measurement of the integrated relaxation pressure of the lower esophageal sphincter (LES), did not differ significantly between the treatment groups (difference, -0.75 mm Hg; 95 % CI: -2.26 to 0.76), nor did improvement in the score on the Gastrointestinal Quality of Life Index (difference, 0.14 points; 95 % CI: -4.01 to 4.28). At 3 months, 57 % of patients in the POEM group and 20 % of patients in the LHM group had reflux esophagitis, as assessed by endoscopy; at 24 months, the corresponding percentages were 44 % and 29 %. The authors concluded that POEM was non-inferior to LHM plus Dor's fundoplication in controlling symptoms of achalasia at 2 years; however, GER was more common among patients who underwent POEM than among those who underwent LHM.

The authors stated that this study had several drawbacks. Less than 50 % of the eligible patients participated in the trial, mainly because of a lack of patient consent to undergo randomization. The surgeons were more experienced in performing LHM plus Dor's fundoplication than the endoscopists were in performing POEM. These researchers did not analyze treatment effects on post-operative pain or on the use of pain medications; in general, the findings at the 2-year follow-up in this study suggested that there was no between-group difference in improvements in patient-reported quality of life (QOL). Patients and trial personnel were aware of the treatment-group assignments because blinding was not possible. This was a potential source of bias given that the primary endpoint was based on patients' reports of symptoms; however, objective assessment by manometry corroborated the primary finding.

Vaezi et al (2020) stated that achalasia is an esophageal motility disorder characterized by aberrant peristalsis and insufficient relaxation of the lower esophageal sphincter (LES). Patients most commonly present with dysphagia to solids and liquids, regurgitation, and occasional chest pain with or without weight loss. High-resolution manometry has identified 3 subtypes of achalasia distinguished by pressurization and contraction patterns. Endoscopic findings of retained saliva with puckering of the gastro-esophageal junction or esophagram findings of a dilated esophagus with bird beaking are important diagnostic clues. In an American College of Gastroenterology (ACG) guideline, the authors used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) process to provide clinical guidance on how best to diagnose and treat patients with achalasia. They stated that "We suggest that POEM or PD result in comparable symptomatic improvement in patients with types I or II achalasia - Low GRADE quality; and "conditional" recommendation".

In a systematic review and network meta-analysis, Facciorusso and colleagues (2021) examined the comparative efficacy of different strategies for management of achalasia. These researchers identified 6 randomized controlled trials (RCTs) in adults with achalasia that compared the efficacy of pneumatic dilation (PD; $n = 260$), laparoscopic Heller myotomy (LHM; $n = 309$), and per-oral endoscopic myotomy (POEM; $n = 176$). Primary efficacy outcome was 1-year treatment success (patient-reported improvement in symptoms based on validated scores); secondary efficacy outcomes were 2-year treatment success and physiologic improvement; safety outcomes were risk of gastro-esophageal reflux disease (GERD), severe erosive esophagitis, and procedure-related serious adverse events (SAEs). These investigators performed pair-wise and network meta-analysis for all treatments, and used GRADE criteria to appraise quality of evidence. Low-quality evidence, based primarily on direct evidence, supports the use of POEM (RR [risk ratio], 1.29; 95 % confidence intervals [CI]: 0.99 to 1.69), and LHM (RR, 1.18 [0.96 to 1.44]) over PD for treatment success at 1 year; no significant difference was observed between LHM and POEM

(RR 1.09 [0.86 to 1.39]). The incidence of severe esophagitis after POEM, LHM, and PD was 5.3 %, 3.7 %, and 1.5 %, respectively. Procedure-related SAE rate after POEM, LHM, and PD was 1.4 %, 6.7 %, and 4.2 %, respectively. The authors concluded that POEM and LHM have comparable efficacy, and may increase treatment success as compared to PD with low confidence in estimates. POEM may have lower rate of SAEs compared to LHM and PD, but higher rate of GERD. Moreover, these researchers stated that future prospective studies comparing long-term safety and efficacy of POEM, LHM, and PD, particularly across specific achalasia subtypes are needed.

The authors stated that this study had several drawbacks. The studies had a short duration of follow-up, and the primary outcome of this network meta-analysis was focused on short-term (1-year) success. Limited post-intervention follow-up prevented the ability to understand long-term comparative efficacy between interventions, which is critical for a chronic disease. There was also a paucity of direct head-to-head comparative trials, in particular comparing POEM to the other treatments. Further, performance and detection bias related to the non-blinded design of included trials introduced significant risk of bias. Performance and detection bias were not easily avoidable in RCTs testing new devices or techniques in surgery or endoscopy, given the nature of the intervention under study, and this represented a limitation in particular when considering subjective outcomes such as improvement of symptoms in patients with achalasia. Assessment of vital physiologic outcomes by blinded readers could overcome limitations related to blinding in these trials; however, these outcomes were infrequently and/or inconsistently reported in trials data, which limited ability to compare post-treatment physiologic efficacy. Similarly, treatment-related AEs were poorly reported and a thorough assessment of risk-benefit profile could not be performed. Finally, inherent to network meta-analyses is risk of misinterpretation due to conceptual heterogeneity, related to differences in subjects, interventions, co-interventions / background treatment, and outcome assessment, which may limit comparability of trials; these could not be adequately accounted for with study-level synthesis, and individual participant-level pooled analyses will be needed.

Cappell et al (2020) reviewed clinical presentation, diagnosis, and therapy of achalasia, focusing on recent developments in high-resolution esophageal manometry (HREM) for diagnosis and POEM for therapy. These researchers carried out a systematic review of achalasia using computerized literature search via PubMed and Ovid of articles published since 2005 with keywords ("achalasia") AND ("high resolution" or "HREM" or "per-oral endoscopic myotomy" or "POEM"). Two authors independently performed literature searches and incorporated articles into this review by consensus according to prospectively determined criteria. Achalasia is an uncommon esophageal motility disorder, usually manifested by dysphagia to solids and liquids, and sometimes manifested by chest pain, regurgitation, and weight loss. Symptoms often suggest more common disorders, such as GERD, thus often delaying diagnosis. Achalasia is a predominantly idiopathic chronic disease. Diagnosis is typically suggested by barium swallow showing esophageal dilation; absent distal esophageal peristalsis; smoothly tapered narrowing ("bird's beak") at esophagogastric junction; and delayed passage of contrast into stomach. Diagnostic findings at (HREM include: distal esophageal aperistalsis and integrated relaxation pressure (through LES pressure during 4 s) of greater than 15 mmHg. Achalasia is classified by HREM into: type 1 classic; type 2 compartmentalized high pressure in esophageal body, and type 3 spastic. This classification impacts therapeutic decisions. Esophagogastroduodenoscopy is required before therapy to assess esophagus and esophagogastric junction and to exclude distal esophageal malignancy. POEM is a revolutionizing achalasia therapy. POEM creates a myotomy via interventional endoscopy. Numerous studies demonstrated that POEM produces comparable, if not superior, results compared to standard LHM, as determined by LES pressure, dysphagia frequency, Eckardt score, hospital length of stay (LOS), therapy durability, and incidence of GERD. Other therapies, including botulinum toxin (BTX) injection and pneumatic dilation, have moderately less efficacy and much less durability than POEM. The authors concluded that this comprehensive review suggested that POEM is equivalent or perhaps superior to LHM for achalasia in terms of cost efficiency, hospital LOS, and relief of dysphagia, with comparable side effects. The data are, however, not conclusive due to sparse long-term follow-up and lack of randomized comparative clinical trials; and POEM therapy is currently limited by a shortage of trained endoscopists.

Martins et al (2020) noted that achalasia is a neurodegenerative motility esophageal disorder characterized by failure of LES relaxation. The conventional therapeutic option for achalasia has been laparoscopic Heller myotomy (LHM). However, in 2010, Inoue et al described POEM, a minimally invasive procedure, as an alternative therapy. To-date, some

studies with small sample sizes have aimed to compare outcomes of LHM versus POEM. These researchers carried out a systematic review and meta-analysis to better-evaluate the safety and efficacy of these 2 procedures. Individualized search strategies were developed from inception through April 2019 in accordance with PRISMA guidelines. Variables analyzed included operative time, overall AEs rate, post-procedure GERD, hospital LOS, post-procedure pain score, and Eckardt Score reduction. A total of 12 cohort trials were selected, consisting of 893 patients (359 in POEM group and 534 in LHM). No randomized clinical trials were available. There was no difference in operative time (MD = -10.26, 95 % CI: -5.6 to 8.2), $p < 0.001$) or Post-Operative Gastroesophageal Reflux (RD: -0.00, 95 % CI: -0.09, to 0.09, I²: 0 %). There was decreased hospital LOS for POEM (MD: -0.6, 95 % CI: -1.11 to -0.09, $p = 0.02$), and an increased mean reduction in Eckardt score in POEM patients (MD = -0.257, 95 % CI: -0.512 to -0.002), $p = 0.048$), with similar rates of AEs. The authors concluded that POEM demonstrated similar results compared to LHM with regards to improvement of dysphagia, post-procedure reflux, and surgical time, with the benefit of shorter hospital LOS; thus, POEM could be considered an option for patients with achalasia. Moreover, these researchers stated that this study had several limitations due to heterogeneity among the studies. Only 1 study had a follow-up period of greater than 12 months, making it impossible to evaluate long-term outcomes. These investigators stated that randomized trials comparing methods are not available yet in the literature, which reduces the level of evidence from this review.

In a systematic review and meta-analysis, Tan et al (2021) examined the safety and efficacy of POEM in achalasia patients with failed previous intervention. These investigators searched the Medline, Embase, Cochrane, and PubMed databases using the queries "achalasia", "per-oral endoscopic myotomy" and related terms in March 2019. Data on technical and clinical success, AEs, Eckardt score and lower esophageal sphincter (LES) pressure were collected. The pooled event rates, mean differences (MDs) and RR were calculated. A total of 15 studies with 2,276 achalasia patients were included. Overall, the pooled technical success, clinical success and AE rate of rescue POEM were 98.0 % (95 % CI: 96.6 % to 98.8 %), 90.8 % (95 % CI: 88.8 % to 92.4 %) and 10.3 % (95 % CI: 6.6 % to 15.8 %), respectively; 7 studies compared the clinical outcomes of POEM between previous failed treatment and the treatment naïve patients. The RR for technical success, clinical success, and AEs were 1.00 (95 % CI: 0.98 to 1.01), 0.98 (95 % CI, 0.92 to 1.04), and 1.17 (95 % CI: 0.78 to 1.76), respectively. Overall, there was significant reduction in the pre- and post-Eckardt score (MD, 5.77; $p < 0.001$) and LES pressure (MD, 18.3 mm Hg; $p < 0.001$) for achalasia patients with failed previous intervention after POEM. The authors concluded that POEM appeared to be a safe, effective and feasible treatment for individuals who had undergone previous failed intervention. It had similar outcomes in previously treated and treatment-naïve achalasia patients. These researchers stated that POEM may be an attractive option for the treatment of patients with this difficult condition; however, further studies with a long-term follow-up to determine the durability of rescue POEM are still needed.

The authors stated that this study had several drawbacks. First, only retrospective and prospective studies were included. No RCTs were found. Second, owing to the paucity of data in the included studies, these researchers were unable to evaluate the safety and efficacy of POEM for patients with previous surgical or endoscopic interventions separately. Third, some studies included pediatric patients with achalasia and some patients with other esophageal dysmotility disorders. However, these patients accounted for a small percentage and the outcomes were unchanged after removing these studies. Fourth, these investigators were unable to examine the quality of life (QOL) in patients with prior treatments after POEM due to the limited number of studies reporting these findings. Last, long-term (greater than 2 years) differences between the patients with and without prior intervention should be interpreted carefully as only 2 studies reported these outcomes.

Dirks et al (2021) examined the literature comparing POEM to Heller myotomy (HM) and pneumatic dilation (PD) for the treatment of achalasia. These researchers carried out a systematic review of comparative studies between POEM and HM or PD. A priori outcomes pertained to efficacy, perioperative metrics, and safety. Internal validity of observational studies and randomized controlled trials (RCTs) was judged using the Newcastle Ottawa Scale and the Cochrane Risk of Bias 2.0 tool, respectively. From 1379 unique literature citations, these investigators included 28 studies comparing POEM and HM ($n = 21$) or PD ($n = 8$), with only 1 RCT addressing each. Aside from two 4-year observational studies, POEM follow-up averaged less than or equal to 2 years. While POEM had similar efficacy to HM, POEM treated dysphagia better than PD both in an RCT

(treatment "success" RR 1.71, 95 % CI: 1.34 to 2.17; 126 patients) and in observational studies (Eckardt score MD - 0.43, 95 % CI: - 0.71 to - 0.16; 5 studies; I2 21 %; 405 patients). POEM needed re-intervention less than PD in an RCT (RR 0.19, 95 % CI: 0.08 to 0.47; 126 patients) and HM in an observational study (RR 0.33, 95 % CI: 0.16 to 0.68; 98 patients). Though 6 to 12 months patient-reported reflux was worse than PD in 3 observational studies (RR 2.67, 95 % CI: 1.02 to 7.00; I2 0 %; 164 patients), post-intervention reflux was inconsistently measured and not statistically different in measures greater than or equal to 1 year. POEM had similar safety outcomes to both HM and PD, including treatment-related serious AEs. The authors concluded that POEM has similar outcomes to HM and greater efficacy than PD; however, reflux remains a critical outcome with unknown long-term clinical significance due to insufficient data and inconsistent reporting.

Zhong et al (2021) stated that POEM is a novel minimally invasive intervention in treating achalasia in adults. Presently, POEM was also reported to be effective for achalasia in children. These researchers examined the clinical outcomes of POEM for pediatric achalasia. They carried out a systematic literature search in PubMed, Embase, and Cochrane databases, which covered the period from January 2009 to June 2020. Selecting studies and collecting data was independently by 2 reviewers according to pre-defined criteria. The statistical analysis was conducted using Comprehensive Meta-Analysis software version 2 and Review Manager 5.3. A total of 11 studies with 389 children were identified in the final analysis. Pooled technical success of POEM treatment achalasia was achieved in 385 children (97.4 %; 95 % CI: 94.7 % to 98.7 %), and the pooled clinical success was achieved in 348 children (92.4 %; 95 % CI: 89.0 % to 94.8 %). After POEM, the Eckardt score was significantly decreased by 6.76 points (95 % CI: 6.18 to 7.34, $p < 0.00001$), and the LES pressure was significantly reduced by 19.38 mmHg (95 % CI: 17.54 to 21.22, $p < 0.00001$). The pooled major AEs rate related to POEM was 12.8 % (95 % CI: 4.5 % to 31.5 %) and the GER rate was 17.8 % (95 % CI: 14.2 % to 22.0 %). The authors concluded that the current study demonstrated that the POEM was a safe and effective technique for treating achalasia in children. Moreover, they stated that further randomized comparative studies of POEM and other therapeutic methods are needed to determine the most effective treatment modality for achalasia in children.

Furthermore, the current version of UpToDate review on "Overview of the treatment of achalasia" (Spechler, 2021) states that "POEM is an effective submucosal endoscopic technique for performing myotomy of the LES and more proximal esophageal muscle. In addition, good results for POEM have been reported in patients with achalasia conditions that often do not respond well to conventional therapies such as type III (spastic) achalasia and "end stage" achalasia (markedly dilated, sigmoid esophagus), and in patients who have failed prior endoscopic and surgical achalasia treatments. The role of POEM in the treatment of achalasia continues to evolve, although there is a consensus that POEM is the procedure of choice for the treatment of type III achalasia. It has been suggested that patients undergoing POEM should be counseled regarding the increased risk of post-procedure reflux compared with other treatments".

Pharyngeal Motor Cortex Stimulation

Michou et al (2012) examined the behavioral and neurophysiological effects of a new neuro-stimulation technique (paired associative stimulation [PAS]), applied to the pharyngeal motor cortex, on swallowing function in healthy individuals and patients with dysphagia from stroke. These researchers examined the optimal parameters of PAS to promote plasticity by combining peripheral pharyngeal (electrical) with cortical stimulation. A virtual lesion was used as an experimental model of stroke, created with 1-Hz repetitive transcranial magnetic stimulation over the pharyngeal cortex in 12 healthy individuals. They tested whether hemispheric targeting of PAS altered swallowing performance before applying the technique to 6 patients with severe, chronic dysphagia from stroke (mean of 38.8 +/- 24.4 weeks post-stroke). Ten minutes of PAS to the unlesioned pharyngeal cortex reversed (bilaterally) the cortical suppression induced by virtual lesion (lesioned: $F(1,9) = 21.347$, $p = 0.001$; contralesional: $F(1,9) = 9.648$, $p = 0.013$; repeated-measures analysis of variance) compared with sham PAS. It promoted changes in behavior responses measured with a swallowing reaction time task ($F(1,7) = 21.02$, $p = 0.003$; repeated-measures analysis of variance). In patients with chronic dysphagia, real PAS induced short-term bilateral changes in the brain; the unaffected pharyngeal cortex had increased excitability ($p = 0.001$; 95 % CI: 0.21 to 0.05; post-hoc paired t test) with reduced penetration-aspiration scores and changes in swallowing biomechanics determined by videofluoroscopy. The authors concluded that the beneficial neurophysiological and behavioral properties of PAS, when applied to unlesioned brain,

provide the foundation for further investigation into the use of neurostimulation as a rehabilitative approach for patients with dysphagia from stroke.

Repetitive Peripheral/Transcranial Magnetic Stimulation

Khedr and Abo-Elfetoh (2010) examined the effect of repetitive transcranial magnetic stimulation (rTMS) applied to the motor area of both hemispheres in patients with acute lateral medullary infarction (LMI) or other brainstem infarctions. The study included 22 patients with acute ischemic stroke who had severe bulbar manifestation. 11 patients had LMI, and 11 had another brainstem infarction. They were randomly allocated to receive active (n = 11) or sham (n = 11) rTMS of the esophageal motor cortex. Each patient received 300 rTMS pulses at 3 Hz and an intensity of 130 % resting motor threshold to each hemisphere for 5 consecutive days. Clinical ratings of dysphagia and motor disability were assessed before and immediately after the last session, and then again after 1 and 2 months. There were no significant differences in baseline clinical assessment of swallowing between active and sham groups. Active rTMS improved dysphagia compared with sham rTMS in both groups of patients, ($p = 0.001$ for both); the LMI group also improved the scores in the Barthel Index. All improvements were maintained over 2 months of follow-up ($p = 0.001$). The authors concluded that these findings suggested that rTMS could be a useful adjuvant strategy in neurorehabilitation of dysphagia due to LMI or other brainstem infarction, although further assessment is needed in multi-center clinical trials.

Momosaki et al (2014) examined the safety and feasibility of a 6-day protocol of bilateral (rTMS combined with intensive swallowing rehabilitation for chronic post-stroke dysphagia. In-hospital treatment was provided to 4 post-stroke patients (age at treatment ranged from 56 to 80 years; interval between onset of stroke and treatment: 24 to 37 months) with dysphagia. Over 6 consecutive days, each patient received 10 sessions of rTMS at 3 Hz applied to the pharyngeal motor cortex bilaterally, followed by 20 mins of intensive swallowing rehabilitation exercise. The swallowing function was evaluated by the Penetration Aspiration Scale (PAS), Modified Mann Assessment of Swallowing Ability (MMASA), Functional Oral Intake Scale (FOIS), laryngeal elevation delay time (LEDT) and Repetitive Saliva-Swallowing Test (RSST) on admission and at discharge. All patients completed the 6-day treatment protocol and none showed any adverse reactions throughout the treatment. The authors concluded that the combination treatment improved laryngeal elevation delay time in all patients. They stated that the proposed protocol of rTMS plus swallowing rehabilitation exercise seemed to be safe and feasible for chronic post-stroke dysphagia, although its effectiveness needs to be confirmed in a large number of patients.

Pisegna and colleagues (2016) evaluated the effects of non-invasive brain stimulation on post-stroke dysphagia. A total of 13 databases were systematically searched through July 2014. Studies had to meet pre-specified inclusion and exclusion criteria. Each study's methodological quality was examined. Effect sizes were calculated from extracted data and combined for an overall summary statistic. A total of 8 RCTs were included. These trials revealed a significant, moderate pooled effect size (0.55; 95 % CI: 0.17 to 0.93; $p = 0.004$). Studies stimulating the affected hemisphere had a combined effect size of 0.46 (95 % CI: -0.18 to 1.11; $p = 0.16$); studies stimulating the unaffected hemisphere had a combined effect size of 0.65 (95 % CI: 0.14 to 1.16; $p = 0.01$). At long-term follow-up, 3 studies demonstrated a large but non-significant pooled effect size (0.81, $p = 0.11$). The authors concluded that this review found evidence for the effectiveness of non-invasive brain stimulation on post-stroke dysphagia. A significant effect size resulted when stimulating the unaffected rather than the affected hemisphere. This finding is in agreement with previous studies implicating the plasticity of cortical neurons in the unaffected hemisphere. The authors stated that non-invasive brain stimulation appeared to assist cortical reorganization in post-stroke dysphagia; however emerging factors high-lighted the need for more data.

Kim and colleagues (2020) examined evidence from published systematic reviews of clinical trials to examine the effectiveness of rTMS in the stroke population. The Cochrane Library, Medline, CINAHL, Embase, and PubMed were searched for systematic reviews up to January 15, 2019. Three authors independently screened the reviews and assessed the methodological quality, using Assessment of Multiple Systematic Reviews (AMSTAR) appraisal tool. Quality of evidence for outcomes evaluated within the reviews was appraised with Grade of Recommendation, Assessment, Development and Evaluation (GRADE) tool. A total of 12 reviews (n = 9,117 subjects) examined the effectiveness of rTMS on motor and non-motor (aphasia, depression, dysphagia and cognition) functions. The rTMS

protocols applied and outcomes measured were diverse among the selected reviews. The findings suggested beneficial effect of rTMS with: "moderate quality" evidence for dysphagia and hemineglect, "low to moderate quality" evidence for motor function (upper limb function, daily activities), and "low quality" evidence for aphasia and post-stroke depression. The authors concluded that despite widespread use of rTMS, high-quality evidence for its routine use for the treatment of stroke survivors is lacking. These researchers stated that further studies are needed to establish differential roles of various protocols and long-term effects of rTMS in the stroke population.

Xie et al (2022) stated that there is no reliable evidence-based medicine to support the effect of rTMS on post-stroke dysphagia. In a meta-analysis, these researchers examined the safety and effectiveness of rTMS on post-stroke dysphagia. English-language literature published before December 20, 2021, were searched in 6 electronic databases. Identified articles were screened, data were extracted, and the methodological quality of included trials was assessed. Meta-analysis was performed using RevMan 5.3 software.

The GRADE method was used to assess the quality of the evidence. A total of 10 studies with 246 patients were included. Meta-analysis showed that rTMS significantly improved overall swallowing function (SMD-0.76, 95 % CI: -1.07 to -0.46, $p < 0.0001$, $n = 206$; moderate-quality evidence), Penetration Aspiration Scale (PAS) (MD-1.03, 95 % CI: -1.51 to -0.55, $p < 0.0001$, $n = 161$; low-quality evidence) and Barthel index scale (BI) (MD 23.86, 95 % CI: 12.73 to 34.99, $p < 0.0001$, $n = 136$; moderate-quality evidence).

Subgroup analyses revealed that rTMS targeting the affected hemisphere and targeting both hemispheres significantly enhanced overall swallowing function and reduced aspiration. Low-frequency rTMS significantly enhanced overall swallowing function and reduced aspiration, and there was no significant difference between high-frequency rTMS and control group in reducing aspiration ($p = 0.09$). There was no statistical difference in the drop-out rate (low-quality evidence) and adverse effects (moderate-quality evidence) between the rTMS group and the control group. The authors concluded that the findings of this meta-analysis study suggested that rTMS had a favorable effect on swallowing function in patients with post-stroke dysphagia; however, there were many parameters that could influence the effectiveness (e.g., the frequency and the site of stimulation). These investigators stated that further research on the mechanism of rTMS and the setting of optimal parameters is needed for the development of this novel intervention in clinical practice.

The authors stated that this study had several drawbacks. First, the small sample size ($n = 12$ to 29) of the included studies may limit the statistical power to detect the effects of rTMS on swallowing function in patients with post-stroke dysphagia. Second, there was considerable heterogeneity in the stimulation parameters (e.g., frequency, intensity and pulse) in the included studies; thus, the optimal stimulation parameters for rTMS were unclear. Third, this study only carried out subgroup analyses for frequency and stimulation site; however, there was heterogeneity in the results of some of the subgroup analyses. Furthermore, the effectiveness of rTMS may also be influenced by other parameters (e.g., brain injury severity and time to stroke onset).

ERBE Electrocautery

Siersema (2008) stated that esophageal strictures are a problem commonly encountered in gastroenterological practice and can be caused by malignant or benign lesions. Dysphagia is the symptom experienced by all patients, regardless of whether their strictures are caused by malignant or benign lesions. The methods most frequently used for palliation of malignant esophageal strictures are stent placement (particularly in patients with an expected survival of 3 months or less) and brachytherapy (in patients with a life expectancy of more than 3 months). Brachytherapy has been shown to be beneficial in patients with an expected survival of longer than 3 months with regard to (prolonged) dysphagia improvement, complications and quality of life. The mainstay of benign esophageal stricture treatment is dilation. Although dilation usually results in symptomatic relief, recurrent strictures do occur. In order to predict which types of strictures are most likely to recur, it is important to differentiate between esophageal strictures that are simple (i.e., focal, straight strictures with a diameter that allows endoscope passage) and those that are more complex (i.e., long (greater than 2 cm), tortuous strictures with a narrow diameter). These complex strictures are considered refractory when they cannot be dilated to an adequate diameter. Novel treatment modalities for refractory strictures include temporary stent placement and incisional therapy.

Hordijk and colleagues (2009) stated that benign gastro-esophageal anastomotic strictures

are common and often refractory to treatment. Various endoscopic dilation techniques have been reported, but none of these methods has been proven to be superior. In a randomized, prospective, multi-center study, these researchers compared the safety and effectiveness of dilation of previously untreated anastomotic strictures by using electrocautery incision (EI) and Savary bougienage (SB). A total of 62 patients with an anastomotic stricture after esophago-gastrostomy and dysphagia Atkinson grades II to IV were included. Patients were treated with EI or SB. Objective and subjective results were compared with baseline and 1, 3, and 6 months after the first treatment. Complications of both treatments were noted. Primary end-points after 6 months were the mean number of dilation sessions and success rate (percentage of patients with less than or equal to 5 dilations in 6 months). Study participation ended after 6 months or if dysphagia grades II to IV recurred despite 5 treatment sessions. No complications occurred with both treatments. There was no significant difference between the EI and SB groups in the mean number of dilations (2.9; 95 % CI: 2.7 to 4.1 versus 3.3; 95 % CI: 2.3 to 3.6; $p = 0.46$) or the success rate (80.6 % versus 67.7 %, $p = 0.26$ and 96.2 % versus 80.8 %, $p = 0.19$). The authors concluded that this prospective trial demonstrated that EI of gastro-esophageal anastomotic strictures is a safe therapy and equivalent to SB as a primary therapy. EI can be used as an alternative or additional therapy to SB. Limitations of this trial: In a small study with negative primary end-points, secondary end-points and subgroup analyses are hypothesis generating only.

Alonso-Larraga et al (2011) stated that dysphagia is a common problem after surgical stenosis (5 % to 55 %) and can be refractory to conventional endoscopic treatment in 22 % of cases. It has been proposed that electro-incision is an alternative and effective treatment. These researchers evaluated the effectiveness of electro-incision with the insulation-tipped diathermic Knife-2 (IT-Knife-2) in the treatment of dysphagia produced by surgical anastomotic strictures. Longitudinal and case-series studies from August 2009 to June 2010 were selected for analysis. A total of 8 consecutive patients with anastomotic stricture-associated dysphagia and naive to endoscopic treatment were included. These investigators performed 3 or more radiated cuts in the stricture until passage of the gastroscope was achieved with IT-Knife-2 and electrocautery (ERBE IC 200) with a 70-100 W energy cut-off and 25 W coagulation. These researchers carried out measurements at baseline and 15 days after the intervention, evaluating the dysphagia by the Atkinson grading scale and endoscopic changes. The majority of patients were at clinical stage IV with an Eastern Cooperative Oncology Group score of 1 to 3 and Karnofsky between 40 and 90. At the time of endoscopic diagnosis, patients had dysphagia grade II and III. Strictures in all of the cases were short in length and had a diameter of minor than 5 mm. At 15 days of the intervention, no patient demonstrated dysphagia ($p = 0.0013$) and the anastomotic diameters was more than 9.5 mm and without evidence of stenosis ($p = 0.0001$). None of the patients presented post-incisional complications. The authors concluded that electro-incision with IT-Knife-2 is effective as primary treatment for the relief of benign dysphagia associated with post-surgical anastomotic stenosis. This was a small study ($n = 8$) with short-term follow-up (15 days). These preliminary findings need to be validated by well-designed studies.

Also, an UpToDate review on "Oropharyngeal dysphagia: Clinical features, diagnosis, and management" (Lembo, 2015) does not mention electrocautery as a therapeutic option.

Palliative Radiotherapy (Including Intra-Cavitary Radiotherapy and Intra-Luminal Brachytherapy)

Adamson et al (2014) noted that the single most distressing symptom for patients with advanced esophageal cancer is dysphagia. Among the more effective treatments for relief of dysphagia is insertion of a self-expanding metal stent (SEMS). It is possible that the addition of a palliative dose of external beam radiotherapy may prolong the relief of dysphagia and provide additional survival benefit. The ROCS (Radiotherapy after Oesophageal Cancer Stenting) trial will assess the effect of adding palliative radiotherapy after esophageal stent insertion. The study is a randomized multi-center phase III trial, with an internal pilot phase, comparing stent alone versus stent plus palliative radiotherapy in patients with incurable esophageal cancer. Eligible participants are those with advanced esophageal cancer who are in need of stent insertion for primary management of dysphagia. Radiotherapy will be administered as 20 Gray (Gy) in 5 fractions over 1 week or 30 Gy in 10 fractions over 2 weeks, within 4 weeks of stent insertion. The internal pilot will assess rates and methods of recruitment; pre-defined criteria will determine progression to the main trial. In total, 496 patients will be randomized in a 1:1 ratio with follow up until death. The primary outcome is time to

progression of patient-reported dysphagia. Secondary outcomes include survival, toxicity, health resource utilization, and quality of life. An embedded qualitative study will explore the feasibility of patient recruitment by examining patients' motivations for involvement and their experiences of consent and recruitment, including reasons for not consenting. It will also explore patients' experiences of each trial arm. These investigators stated that the ROCS study will be a challenging trial studying palliation in patients with a poor prognosis. The internal pilot design will optimize methods for recruitment and data collection to ensure that the main trial is completed on time. As a pragmatic trial, study strengths include collection of all follow-up data in the usual place of care, and a focus on patient-reported, rather than disease-orientated, outcomes. Exploration of patient experience and health economic analyses will be integral to the assessment of benefit for patients and the National Health Service (NHS). The trial was registered with Current Controlled Trials (registration number: ISRCTN12376468) on July 10, 2012.

In a Cochrane review, Dai and associates (2014) analyzed the effectiveness of different interventions used in the palliation of dysphagia in primary esophageal and gastro-esophageal carcinoma. In this updated review (January 2014), these investigators searched, according to the Cochrane Upper Gastrointestinal and Pancreatic Diseases model, the Cochrane Central Register of Controlled Trials (CENTRAL) (The Cochrane Library), MEDLINE, EMBASE and CINAHL; and major conference proceedings (up to January 2014). Only RCTs were included in which patients with inoperable or unresectable primary esophageal cancer underwent palliative treatment. Different interventions like rigid plastic intubation, SEMS insertion, brachytherapy, external beam radiotherapy, chemotherapy, esophageal bypass surgery, chemical and thermal ablation therapy, either head-to-head or in combination, were included. The primary outcome was dysphagia improvement. Secondary outcomes included recurrent dysphagia, technical success, procedure related mortality, 30-day mortality, adverse effects and quality of life. Data collection and analysis were performed in accordance with the methods of the Cochrane Upper Gastrointestinal and Pancreatic Diseases Review Group. These researchers included 3,684 patients from 53 studies. Self-expanding metallic stent insertion was safer and more effective than plastic tube insertion. Thermal and chemical ablative therapy provided comparable dysphagia palliation but had an increased requirement for re-interventions and for adverse effects. Anti-reflux stents provided comparable dysphagia palliation to conventional metal stents. Some anti-reflux stents might have reduced gastro-esophageal reflux and complications. Newly-designed double-layered nitinol (Niti-S) stents were preferable due to longer survival time and fewer complications compared to simple Niti-S stents. Brachytherapy might be a suitable alternative to SEMS in providing a survival advantage and possibly a better quality of life, and might provide better results when combined with argon plasma coagulation or external beam radiation therapy. The authors concluded that SEMS insertion is safe, effective and quicker in palliating dysphagia compared to other modalities. However, high-dose intra-luminal brachytherapy is a suitable alternative and might provide additional survival benefit with a better quality of life. Some anti-reflux stents and newly-designed stents lead to longer survival and fewer complications compared to conventional stents. Combinations of brachytherapy with SEMS insertion or radiotherapy are preferable due to the reduced requirement for re-interventions. Rigid plastic tube insertion, dilatation alone or in combination with other modalities, and chemotherapy alone are not recommended for palliation of dysphagia due to a high incidence of delayed complications and recurrent dysphagia.

Yamashita and colleagues (2015) stated that intra-cavitary radiotherapy (ICRT) for the palliative treatment of advanced esophageal cancer with dysphagia is currently performed at the University of Tokyo Hospital (Tokyo, Japan). In the present study, 24 patients exhibiting advanced esophageal cancer with dysphagia received palliative ICRT, which was delivered 5 mm below the esophageal mucous membrane, with the exception of 1 case, was administered at a dose of 6 Gy/fraction. Specific patients additionally underwent definitive or palliative external beam radiation therapy for esophageal cancer a minimum of 3 months prior to ICRT. The effect of treatment on symptom alleviation was examined by comparing the dysphagia score prior to and following ICRT, with the patients' medical records and a questionnaire used to calculate a dysphagia score ranging from 0 (no dysphagia) to 4 (total dysphagia). In consideration of the individual efficacy of the treatment, the maximum number of repeated ICRT fractions was 4 (median of 1.7 times). A trend in the improvement of the symptom of dysphagia was observed in response to esophageal ICRT, with the average dysphagia score markedly decreasing from 2.54 to 1.65, however, the difference was not significant ($p = 0.083$). Furthermore, pain was the most frequent side-effect of the esophageal ICRT and no patients exhibited severe complications. The authors concluded that esophageal ICRT at a dose of 6 Gy/fraction may

present an effective strategy for relieving the symptom of dysphagia in cases of advanced esophageal cancer. These preliminary findings need to be validated by well-designed studies.

The Intensive Dysphagia Rehabilitation Approach (e.g., the Swallow STRONG Multi-Disciplinary Oropharyngeal Strengthening Program)

Rogus-Pulia et al (2016) noted that dysphagia is associated with malnutrition, aspiration pneumonia, and mortality in older adults. Strengthening interventions have shown promising results, but the effectiveness of treating dysphagia in older adults remains to be established. The Swallow STRengthening OropharyNGeal (Swallow STRONG) Program is a multi-disciplinary program that employs a specific approach to oropharyngeal strengthening, device-facilitated (D-F) isometric progressive resistance oropharyngeal (I-PRO) therapy, with the goal of reducing health-related sequelae in veterans with dysphagia. Participants completed 8 weeks of D-F I-PRO therapy while receiving nutritional counseling and respiratory status monitoring. Assessments were completed at baseline, 4, and 8 weeks. At each visit, video-fluoroscopic swallowing studies were carried out. Dietary and swallowing-related quality of life questionnaires were administered. Long-term monitoring for 6 to 17 months after enrollment allowed for comparison of pneumonia incidence and hospitalizations to the 6 to 17 months before the program. Veterans with dysphagia confirmed with videofluoroscopy ($n = 56$; 55 males, 1 female; mean age of 70 years) were enrolled. Lingual pressures increased at anterior (effect estimate = 92.5, $p < 0.001$) and posterior locations (effect estimate = 85.4, $p < 0.001$) over 8 weeks. Statistically significant improvements occurred on 8 of 11 subscales of the Quality of Life in Swallowing Disorders (SWAL-QOL) Questionnaire (effect estimates = 6.5 to 19.5, $p < 0.04$) and in self-reported sense of effort (effect estimate = -18.1, $p = 0.001$). Higher Functional Oral Intake Scale scores (effect estimate = 0.4, $p = 0.02$) indicated that participants were able to eat less-restrictive diets. There was a 67 % reduction in pneumonia diagnoses, although the difference was not statistically significant. The number of hospital admissions decreased significantly (effect estimate = 0.96; $p = 0.009$) from before to after enrollment. The authors concluded that the findings of this study suggested that the Swallow STRONG multi-disciplinary oropharyngeal strengthening program may be an effective treatment for older adults with dysphagia. These preliminary findings need to be validated by well-designed studies.

In an intervention study (before-after trial) with 4-week follow-up through an online survey, Malandraki and colleagues (2016) examined the effects of the Intensive Dysphagia Rehabilitation approach on physiological and functional swallowing outcomes in adults with neurogenic dysphagia. A consecutive sample of subjects ($n = 10$) recruited from outpatient university clinics were included in this analysis. All subjects were diagnosed with adult-onset neurologic injury or disease. Dysphagia diagnosis was confirmed through clinical and endoscopic swallowing evaluations. No subjects withdrew from the study. Participants completed the 4-week Intensive Dysphagia Rehabilitation protocol, including 2 oropharyngeal exercise regimens, a targeted swallowing routine using salient stimuli, and caregiver participation. Treatment included hourly sessions twice-weekly and home practice for approximately 45 minutes/day. Outcome measures evaluating pre- and post-treatment included airway safety using an 8-point Penetration Aspiration Scale, lingual isometric pressures, self-reported swallowing-related QOL, and level of oral intake. Also, patients were monitored for adverse dysphagia-related effects; QOL and adverse effects were also assessed at the 4-week follow-up (online survey). The Intensive Dysphagia Rehabilitation approach was effective in improving maximum and mean Penetration Aspiration Scale scores ($p < 0.05$, $\eta(2) = 0.8146$ and $p < 0.05$, $\eta(2) = 0.799708$, respectively) and level of oral intake ($p < 0.005$, Cohen $d = -1.387$). Of the 5 patients who were feeding tube-dependent initially, 2 progressed to total oral nutrition, and 2 progressed to partial oral nutrition; 1 patient remained tube-dependent; QOL was significantly improved at the 4-week follow-up (95 % CI: 6.38 to 14.5; $p < 0.00$), but not at the post-treatment. No adverse effects were observed/ reported. The authors concluded that the Intensive Dysphagia Rehabilitation approach was safe and improved physiological and some functional swallowing outcomes in this sample; however, further investigation is needed before it can be widely applied.

Progressive Muscle Diseases

In a Cochrane review, Jones and colleagues (2016) examined the effects of interventions for dysphagia in people with long-term, progressive muscle disease. On January 11, 2016, these investigators searched the Cochrane Neuromuscular Specialized Register, the

Cochrane Central Register of Controlled Trials (CENTRAL), Medline, Embase, AMED, LILACS, and CINAHL. They checked references in the identified trials for additional RCTs and quasi-RCTs. They also searched ClinicalTrials.gov and the World Health Organization International Clinical Trials Registry Platform on January 12, 2016 for ongoing or completed but unpublished clinical trials. These researchers included RCTs and quasi-RCTs that evaluated the effect of interventions for managing dysphagia in adults and children with long-term, progressive muscle disease, compared to other interventions, placebo, no intervention, or standard care. In this review update, the authors decided to include trials of people with sporadic inclusion body myositis (IBM) on the basis that it presents as a long-term, progressive muscle disease with uncertain degenerative and inflammatory etiology and is typically refractory to treatment. They applied standard Cochrane methodological procedures. There were no RCTs that reported results in terms of the review's primary outcome of interest, weight gain or maintenance. However, these researchers identified 1 RCT that assessed the effect of intravenous immunoglobulin (IVIG) on swallowing function in people with IBM. The trial authors did not specify the number of subjects who had dysphagia. There was also incomplete reporting of findings from video-fluoroscopic investigations, which was one of the review's secondary outcome measures. The study did report reductions in the time taken to swallow, as measured using ultrasound. No serious adverse events (AEs) occurred during the study, although data for the follow-up period were lacking. It was also unclear whether the non-serious AEs reported occurred in the treatment group or the placebo group. These investigators assessed this study as having a high risk of bias and uncertain CIs for the review outcomes, which limited the overall quality of the evidence. Using GRADE criteria, the authors down-graded the quality of the evidence from this RCT to "low" for efficacy in treating dysphagia, due to limitations in study design and implementation, and indirectness in terms of the population and outcome measures. Similarly, they assessed the quality of the evidence for AEs as "low". From their search for RCTs, these researchers identified 2 other non-randomized studies, which reported the effects of long-term IVIG therapy in adults with IBM and lip-strengthening exercises in children with myotonic dystrophy type 1. Headaches affected 2 participants treated with long-term IVIG therapy, who received a tailored dose reduction; there were no AEs associated with lip-strengthening exercises. Both non-randomized studies identified improved outcomes for some participants following the intervention, but neither study specified the number of participants with dysphagia or demonstrated any group-level treatment effect for swallowing function using the outcomes pre-specified in this review. The authors concluded that there is insufficient and low-quality RCT evidence to determine the effect of interventions for dysphagia in long-term, progressive muscle disease. Clinically relevant effects of IVIG for dysphagia in patients with IBM can neither be confirmed or excluded using the evidence presented in this review. They stated that standardized, validated, and reliable outcome measures are needed to assess dysphagia and any possible treatment effect. Clinically meaningful outcomes for dysphagia may require a shift in focus from measures of impairment to disability associated with oral feeding difficulties.

Combined Behavioral Treatment with Valproic Acid

Kim and colleagues (2018) noted that palatal myoclonus (PM) is a rare disease that may induce dysphagia. Since dysphagia related to PM is unique and is characterized by myoclonic movements of the involved muscles, specific treatments are needed for rehabilitation. However, no study has investigated the treatment effectiveness for this condition. In a case-report study, these investigators described the benefit of combining behavioral treatment with valproic acid administration in patients with dysphagia triggered by PM. The 2 cases were treated with the combined treatment. The outcomes evaluated by video-fluoroscopic swallowing studies (VFSS) before and after the treatment showed significant decreases in myoclonic movements and improved swallowing function. The authors concluded that the combined treatment yielded positive effects in both cases. The follow-up VFSS showed a clear reduction in both amplitude and frequency of myoclonic movements, and the swallowing function improved substantially. They suggested that valproic acid is effective in reducing myoclonic movements, and behavioral treatment adequately restored the functional ability of the affected muscles in this condition. These preliminary findings need to be validated by well-designed studies.

Laryngeal Manipulation

In a retrospective, single-center, case-series study, DePietro and associates (2018) described the use of laryngeal manipulation as a treatment for dysphagia resulting from

excessive para-laryngeal muscle tension and identified the patients likely to have symptomatic improvement. This trial identified patients from 2007 to 2013 with laryngeal manipulation for muscle tension dysphagia in an academic otolaryngology practice. Subjects with dysphagia not attributable to an anatomic cause who attended therapy at least 1 time were included in the study. The primary outcome of the study was subjective improvement in dysphagia symptoms (yes/no) during follow-up. Symptoms, demographic information, treatment, and response to therapy were recorded. The Kruskal-Wallis test was used for analysis of continuous variables, while a Chi-squared test or Fisher's exact test was used for analysis of all categorical variables. A total of 44 subjects were included, consisting of 37 women and 7 men. Subjective improvement in dysphagia was seen in 34 subjects (77.3 %). No significant differences were seen in improvement based on gender ($p = 0.3223$), race ($p = 0.4317$), number of sessions with a speech pathologist ($p = 0.3198$), or presenting symptoms including hoarseness ($p = 0.0853$), pain ($p = 1.000$), globus ($p = 0.2834$), and cough ($p = 1.000$). The authors found subjective improvement with laryngeal manipulation as reported during follow-up visits to clinic among individuals with muscle tension dysphagia. Patient age, presenting symptoms, and number of therapy sessions were not found to be significantly associated with resolution of symptoms. This was a retrospective, case-series study with relatively small sample size ($n = 44$). It is also interesting to note that the number of therapy sessions was not significantly associated with resolution of symptoms. These findings need to be validated by well-designed studies.

Pharyngeal Electrical Stimulation

Restivo and Hamdy (2018) stated that neurogenic dysphagia (ND) can occur in patients with nervous system diseases of varying etiologies. Moreover, recovery from ND is not guaranteed. The therapeutic approaches for oropharyngeal ND have drastically changed over past 10 years, mainly due to a better understanding of the neurophysiology of swallowing along with the progress of neuroimaging and neurophysiological studies. Thus, it is a priority to develop a treatment that is safe, repeatable, and can be performed at the bedside as well as for out-patients. Pharyngeal electrical stimulation (PES) is a novel rehabilitation treatment for ND; it is performed via location-specific intraluminal catheters that are introduced trans-nasally and enable clinicians to stimulate the pharynx directly. This technique has demonstrated increasingly promising evidence in improving swallowing performance in patients with ND associated with stroke and multiple sclerosis (MS), probably by increasing the cortico-bulbar excitability and inducing cortical re-organization of swallowing motor cortex. The authors concluded that in comparison with other techniques, PES has been demonstrated to be a promising therapeutic option for dysphagia in stroke and MS patients. Although in a recent large trial of dysphagic stroke patients, PES was unable to demonstrate superiority on radiological aspiration or on clinical dysphagia, several factors including under-treatment of patients receiving PES may have contributed to these results. Indeed, the above results differ from the findings of a previous meta-analysis of 3 smaller RCTs, which showed an effect at 2 weeks compared to sham. They stated that further studies selecting stroke patients with more severe dysphagia or those requiring intensive care ventilation need to be designed. In addition, studies evaluating the effect of PES on dysphagia due to different neurological disorders, as well as the association of PES in combination with other rehabilitative treatments, are needed.

Transcranial Direct Current Stimulation

In a double-blind, randomized study, Suntrup-Krueger and colleagues (2018) examined if transcranial direct current stimulation (tDCS) is able to enhance dysphagia rehabilitation following stroke. Besides relating clinical effects with neuroplastic changes in cortical swallowing processing, these researchers identified factors influencing treatment success. A total of 60 acute dysphagic stroke patients received contralesional anodal (1mA, 20 minutes) or sham tDCS on 4 consecutive days. Swallowing function was thoroughly assessed before and after the intervention using the validated fiberoptic endoscopic dysphagia severity scale (FEDSS) and clinical assessment. In 10 patients, swallowing-related brain activation was recorded applying magnetoencephalography before and after the intervention. Voxel-based statistical lesion pattern analysis was also performed. Study groups did not differ according to demographic data, stroke characteristics, or baseline dysphagia severity. Patients treated with tDCS showed greater improvement in FEDSS than the sham group (1.3 versus 0.4 points, mean difference [MD] = 0.9, 95 % CI: 0.4 to 1.4, $p < 0.0005$). Functional recovery was accompanied by a significant increase of activation ($p < 0.05$) in the

contralesional swallowing network after real but not sham tDCS. Regarding predictors of treatment success, for every hour earlier that treatment was initiated, there was greater improvement on the FEDSS (adjusted OR = 0.99, 95 % CI: 0.98 to 1.00, $p < 0.05$) in multivariate analysis. Stroke location in the right insula and operculum was indicative of worse response to tDCS ($p < 0.05$). The authors concluded that application of tDCS over the contralesional swallowing motor cortex supported swallowing network re-organization, thereby leading to faster rehabilitation of acute post-stroke dysphagia. Moreover, they noted that early treatment initiation appeared beneficial; tDCS may be less effective in right-hemispheric insulo-opercular stroke.

In a systematic review and network meta-analysis, Li et al (2021) examined the effect of non-invasive brain stimulation (NIBS) on dysphagia after stroke and compared the effects of 2 different types of NIBS. These researchers retrieved RCTs on the effect of NIBS on dysphagia after stroke from databases of PubMed, Embase, Cochrane Library, Web of Science, CNKI, Wanfang Data, VIP, and CBM, from inception to June 2021. The quality of the trials was assessed, and the data were extracted according to the Cochrane Handbook for Systematic Reviews of Interventions. A statistical analysis was performed using RevMan 5.3 and ADDIS 1.16.8. The effect size was evaluated by using the standardized mean difference (SMD) and a 95 % CI. A total of 18 studies involving 738 patients were included. Meta-analysis showed that NIBS could improve the dysphagia outcome and severity scale (DOSS) score (SMD = 1.44, 95 % CI: 0.80 to 2.08, $p < 0.05$) and the water swallow test score (SMD = 6.23, 95 % CI: 5.44 to 7.03, $p < 0.05$). NIBS could reduce the standardized swallowing assessment (SSA) score (SMD = -1.04, 95 % CI: -1.50 to -0.58, $p < 0.05$), the penetration-aspiration scale (PAS) score (SMD = -0.85, 95 % CI: -1.33 to -0.36, $p < 0.05$), and the functional dysphagia scale score (SMD = -1.05, 95 % CI: -1.48 to -0.62, $p < 0.05$). Network meta-analysis showed that the best probabilistic ranking of the effects of 2 different NIBS on the DOSS score was rTMS ($p = 0.52$) > tDCS ($p = 0.48$), the best probabilistic ranking of the SSA score was rTMS ($p = 0.72$) > tDCS ($p = 0.28$), and the best probabilistic ranking of the PAS score was rTMS ($p = 0.68$) > tDCS ($p = 0.32$). The authors concluded that existing evidence showed that NIBS could improve swallowing dysfunction and reduce the occurrence of aspiration following stroke, and that rTMS was better than tDCS. Moreover, these researchers stated that limited by the number of included studies, more large-sample, multi-center, double-blind, high-quality RCTs are still needed to confirm the findings of this research.

These researchers stated that there are relatively few studies on the mechanism of NIBS improving dysphagia after stroke, and there is no clear conclusion at present. Moreover, they noted that although this study followed the criteria of systematic review and network meta-analysis report (PRISMA statement), there were certain potential biases. Currently, there are relatively few RCTs regarding NIBS to improve dysphagia after stroke. The included studies were published in both Chinese and English, and the lack of relevant gray studies may lead to the bias of study selection. Some studies did not describe specific random methods, allocation concealment, or blind methods. Only 7 studies reported follow-up, the follow-up time was short, and there was no long-term effect of NIBS on the swallowing function in stroke patients. The intervention time was 1 to 8 weeks, and the stimulation sites were unaffected, affected, or bilateral. Furthermore, the frequency and current of stimulation were also different. This study could not determine the relationship between the length of the intervention period, the stimulation site and the intensity of stimulation, and the improvement of swallowing function.

Cenci et al (2024) noted that swallowing is a complex function that can be disrupted following stroke; tDCS is a non-invasive brain stimulation therapy that recently has been tested for the treatment of stroke-related dysphagia. These investigators carried out a search in the literature to examine the described evidence of the use of tDCS in dysphagia after stroke; 3 electronic databases were searched. The risk of bias evaluation was performed using the RoB-2 tool. The GRADE frame-work was also implemented. Of 265 studies, only 9 were included in this review. The most common location of the tDCS stimulation was the unaffected hemisphere (44 %). Regarding the outcome measure, the DOSS was the most commonly used (55 %). However, due to the high heterogeneity of the protocols, and considering the differences between the types of stroke, these researchers opted not to conduct a meta-analysis. Instead, a systematic review with a thorough analysis of each individual study and the impact of the differences to the outcomes was preferred. The authors concluded that the final considerations were that even though the majority of studies described benefits from tDCS in post-stroke dysphagia, as they presented too many methodological differences, it was not possible to compare them. Furthermore, many studies included patients with less than

6 months after stroke, which is an important bias as the swallowing function can recover spontaneously within this period, turning the certainty of the evidence really low.

Swallowing Therapy With or Without Biofeedback

Bath and associates (2018) noted that dysphagia, which is common in patients following stroke, is associated with increased risk of death or dependency, occurrence of pneumonia, poor QOL, and longer hospital stay. Treatments provided to improve dysphagia are aimed at accelerating recovery of swallowing function and reducing these risks. In an update of previous Cochrane reviews (first published in 1999 and updated in 2012), these investigators examined the effects of swallowing therapy on death or dependency among stroke survivors with dysphagia within 6 months of stroke onset. They searched the Cochrane Stroke Group Trials Register (June 26, 2018), the Cochrane Central Register of Controlled Trials (CENTRAL; 2018, Issue 6) in the Cochrane Library (searched June 26, 2018), Medline (June 26, 2018), Embase (June 26, 2018), the Cumulative Index to Nursing and Allied Health Literature (CINAHL) (June 26, 2018), Web of Science Core Collection (June 26, 2018), SpeechBITE (June 28, 2016), ClinicalTrials.gov (June 26, 2018), and the World Health Organization International Clinical Trials Registry Platform (June 26, 2018). They also searched Google Scholar (June 7, 2018) and the reference lists of relevant trials and review articles. These researchers included RCTs of interventions for people with dysphagia and recent stroke (within 6 months). Two review authors independently applied the inclusion criteria, extracted data, assessed risk of bias, used the GRADE approach to assess the quality of evidence, and resolved disagreements through discussion with the 3rd review author. They used random-effects models to calculate ORs, MDs, and standardized MDs (SMDs), and provided 95 % CIs for each. The primary outcome was functional outcome, defined as death or dependency (or death or disability), at the end of the trial. Secondary outcomes were case fatality at the end of the trial, length of inpatient stay, proportion of subjects with dysphagia at the end of the trial, swallowing ability, penetration aspiration score, or pneumonia, pharyngeal transit time, institutionalization, and nutrition. These investigators added 27 new studies (1,777 subjects) to this update to include a total of 41 trials (2,660 subjects). They evaluated the efficacy of swallowing therapy overall and in subgroups by type of intervention: acupuncture (11 studies), behavioral interventions (9 studies), drug therapy (3 studies), neuromuscular electrical stimulation (NMES; 6 studies), pharyngeal electrical stimulation (PES; 4 studies), physical stimulation (3 studies), transcranial direct current stimulation (tDCS; 2 studies), and transcranial magnetic stimulation (TMS; 9 studies). Swallowing therapy had no effect on the primary outcome (death or dependency/disability at the end of the trial) based on data from 1 trial (2 data-sets) (OR 1.05, 95 % CI: 0.63 to 1.75; 306 participants; 2 studies; $I^2 = 0\%$; $p = 0.86$; moderate-quality evidence). Swallowing therapy had no effect on case fatality at the end of the trial (OR 1.00, 95 % CI: 0.66 to 1.52; 766 subjects; 14 studies; $I^2 = 6\%$; $p = 0.99$; moderate-quality evidence). Swallowing therapy probably reduced length of inpatient stay (MD -2.9, 95 % CI: -5.65 to -0.15; 577 subjects; 8 studies; $I^2 = 11\%$; $p = 0.04$; moderate-quality evidence). Researchers found no evidence of a subgroup effect based on testing for subgroup differences ($p = 0.54$). Swallowing therapy may have reduced the proportion of subjects with dysphagia at the end of the trial (OR 0.42, 95 % CI: 0.32 to 0.55; 1,487 participants; 23 studies; $I^2 = 0\%$; $p = 0.00001$; low-quality evidence). Trial results showed no evidence of a subgroup effect based on testing for subgroup differences ($p = 0.91$). Swallowing therapy may improve swallowing ability (SMD -0.66, 95 % CI: -1.01 to -0.32; 1,173 participants; 26 studies; $I^2 = 86\%$; $p = 0.0002$; very low-quality evidence). These researchers found no evidence of a subgroup effect based on testing for subgroup differences ($p = 0.09$). They noted moderate to substantial heterogeneity between trials for these interventions. Swallowing therapy did not reduce the penetration aspiration score (i.e., it did not reduce radiological aspiration) (SMD -0.37, 95 % CI: -0.74 to -0.00; 303 subjects; 11 studies; $I^2 = 46\%$; $p = 0.05$; low-quality evidence). Swallowing therapy may reduce the incidence of chest infection or pneumonia (OR 0.36, 95 % CI: 0.16 to 0.78; 618 subjects; 9 studies; $I^2 = 59\%$; $p = 0.009$; very low-quality evidence). The authors concluded that moderate- and low-quality evidence suggested that swallowing therapy did not have a significant effect on the outcomes of death or dependency/disability, case fatality at the end of the trial, or penetration aspiration score. However, swallowing therapy may have reduced length of hospital stay, dysphagia, and chest infections, and may have improved swallowing ability. However, these researchers stated that these findings were based on evidence of variable quality, involving a variety of interventions; they stated that further high-quality trials are needed to examine if specific interventions are effective.

In a systematic review and meta-analysis, Benfield and colleagues (2019) evaluated the current evidence on the effects of swallow therapy augmented by biofeedback in adults with dysphagia. Two independent reviewers conducted searches that included Medline, Embase, trial registries, and gray literature up to December 2016; RCTs and non-RCTs were assessed, including for risk of bias and quality. Data were extracted by 1 reviewer and verified by another on biofeedback type, measures of swallow function, physiology and clinical outcome, and analyzed using Cochrane Review Manager (random effects models); results were expressed as WMD and OR. Of 675 articles, these researchers included 23 studies (n = 448 subjects); and 3 main types of biofeedback were used: accelerometry, surface electromyography (sEMG), and tongue manometry. Exercises included saliva swallows, maneuvers, and strength exercises. Dose varied between 6 and 72 sessions for 20 to 60 mins; 5 controlled studies (stroke n = 95; head and neck cancer n = 33; mixed etiology n = 10) were included in meta-analyses. Compared to control, biofeedback augmented dysphagia therapy significantly enhanced hyoid displacement (3 studies, WMD = 0.22 cm; 95 % CI: 0.04 to 0.40, p = 0.02); but there was no significant difference in functional oral intake (WMD = 1.10; 95 % CI: -1.69 to 3.89, p = 0.44) or dependency on tube feeding (OR = 3.19; 95 % CI: 0.16 to 62.72, p = 0.45). Risk of bias was high and there was significant statistical heterogeneity between trials in measures of swallow function and number tube fed (I² = 70 % to 94 %). Several non-validated outcome measures were used. Subgroup analyses were not possible due to a paucity of studies. The authors concluded that dysphagia therapy augmented by biofeedback using sEMG and accelerometry enhanced hyoid displacement but functional improvements in swallowing were not evident. These researchers stated that available data are extremely limited and further larger well-designed RCTs are needed.

Duncan and colleagues (2020) examined the effectiveness of dysphagia interventions compared to standard care in improving oral intake and reducing aspiration for adults in acute and critical care. These investigators searched electronic literature for randomized and quasi-randomized trials and bibliography lists of included studies to March 2020. Study screening, data extraction, risk of bias and quality assessments were conducted independently by 2 reviewers. Meta-analysis used fixed effects modelling. These researchers identified 22 studies (19 stroke, 2 intensive care stroke and 1 general intensive care) testing 9 interventions and representing 1,700 patients. Swallowing treatment showed no evidence of a difference in the time to return to oral intake (n = 33, MD (days) - 4.5, 95 % CI: - 10.6 to 1.6, 1 study, p = 0.15) (very low certainty) or in aspiration following treatment (n = 113, RR 0.79, 95 % CI: 0.44 to 1.45, 4 studies, I² = 0 %, p = 0.45) (low certainty). Swallowing treatment showed evidence of a reduced risk of pneumonia (n = 719, RR 0.71, 95 % CI: 0.56 to 0.89, 8 studies, I² = 15 %, p = 0.004) (low certainty); but no evidence of a difference in swallowing QOL scores (n = 239, MD - 11.38, 95 % CI: - 23.83 to 1.08, I² = 78 %, p = 0.07) (very low certainty). The authors concluded that there is limited evidence for the effectiveness of swallowing treatments in the acute and critical care setting; clinical trials consistently measuring patient-centered outcomes are needed.

Telerehabilitation

Nordio and colleagues (2018) stated that some recent studies have shown positive effects of tele-rehabilitation of swallowing disorders, yet there are no systematic reviews verifying the evidence. In a systematic review, these researchers examined the effects of tele-rehabilitation in the field of dysphagia as an alternative to face-to-face patient care, considering swallowing recovery and/or QOL in different patient populations. They searched the Cochrane Library, Medline, Embase, Google Scholar, Google Search and the grey literature from inception until December 2016 for publications written in English (keywords: tele-rehabilitation, telemedicine, dysphagia, swallowing disorders), which resulted in 330 records. Abstract screening and data extraction was performed independently by 2 reviewers; 4 papers were selected to be read in full, and the methodological quality of the studies included was evaluated using Cochrane Collaboration's tool for assessing risk of bias. One study met the inclusion criteria, which showed that tele-rehabilitation improved adherence to treatment compared to patient-directed intervention. Although adherence is an important factor that influences the treatment outcome, clinical outcomes had to be examined in RCTs in order to reach evidence in this field. Lastly, this systematic review did not demonstrate the efficacy of tele-rehabilitation compared with face-to face therapy. These researchers stated that additional research in the field of dysphagia is needed to define the benefits and risks associated with assisting patients in tele-rehabilitation modality. They noted that future trials need to have a more robust design and evaluate the effectiveness of

treatments based on specific clinical outcomes and QOL, in different populations of patients with different type and severity of dysphagia, and associated with cost-accountability and cost-effectiveness analysis.

The authors stated that this review had several drawbacks. First, most of the studies in tele-rehabilitation concerning dysphagia are characterized by small sample sizes, large variability of illnesses included, heterogeneity of treatments and outcomes and the consistent presence of biases. The drawback of small samples was common in the rehabilitation literature concerning dysphagia, because of the different diagnoses that can cause it, the large range of severity and disability experienced by patients and the long time needed to observe changes in clinical outcomes. In addition, these investigators chose to distinguish tele-rehabilitation from other telemedicine applications (e.g., telemonitoring, teleradiology) because of the possibility of providing therapeutic interventions, remotely controlled by healthcare professionals, with rehabilitation intent. Moreover, the authors only considered trials reported in English, which restricted the raw dataset of records used for screening.

Non-Pharmacological, Non-Surgical Treatments for Dysphagia

Eskildsen et al (2021) noted that dysphagia is a common and critical consequence of acquired brain injury (ABI) and could result in severe complications. Dysphagia rehabilitation is transforming from mainly compensatory strategies to the retraining of swallowing function using principles from neuroscience; however, there are no studies that map interventions available to retrain swallowing function in patients with moderate-to-severe ABI. These researchers systematically mapped the accessible scientific literature to address the research question: Which non-surgical, non-pharmacological interventions are used in the treatment of dysphagia in individuals with moderate and severe ABI in the acute and subacute phase? They carried out a scoping review based on the methodology of Arksey and O'Malley and methodological advancement by Levac et al. Data sources included Medline, Embase, Cochrane Library, CINAHL, PsycINFO, Web of Science, OTseeker, speechBITE and PEDro; they were searched up until March 14, 2021. All studies reporting rehabilitative interventions within 6 months of injury for patients with moderate-to-severe ABI and dysphagia were included. Data were extracted by 2 independent reviewers and studies were categorized based on treatment modality. A total of 21,396 records were retrieved, and a final of 26 studies were included.

Interventions were categorized into cortical or non-cortical stimulation of the swallowing network. Cortical stimulation interventions were rTMS and tDCS. Non-cortical interventions were complex swallowing interventions, neuromuscular electrical stimulation, pharyngeal electrical stimulation (PES), sensory stimulation, strengthening exercises and respiratory muscle training. The authors concluded that this scoping review provided an overview of rehabilitative dysphagia interventions for patients with moderate and severe ABI, predominantly due to stroke, in the acute and subacute phase.

Positive tendencies towards beneficial effects were found for rTMS, complex swallowing interventions, PES and cervical strengthening. Moreover, these researchers stated that future studies could benefit from clear reporting of patient diagnosis and disease severity, the use of more standardized treatment protocols or algorithms and fewer but standardized outcome measures to enable comparison of effects across studies and interventions.

The authors stated that this review had several drawbacks. First, missing data on brain injury severity in several studies led to excluding some possibly relevant studies. This information was often unavailable from the corresponding authors. Second, the timing of the assessment of brain injury severity was often not reported or consistent between studies, making it difficult to determine whether the study met the inclusion criteria.

Third, the limitations due to necessary language restrictions caused the exclusion of Asian language papers, potentially excluding some relevant studies. Fourth, given that the included studies have not been quality evaluated, the summarization of results should be interpreted with caution and could not be directly applied to guide clinical practice.

International best practice consensus recommendations for dysphagia management (Jesus Oliviera et al, 2021) state that "[i]t is recommended to consider behavioral intervention, including compensatory strategies (modification of food consistencies and liquid viscosity, postural and swallowing techniques) and rehabilitation (muscle-strengthening exercises, resistance or skills training), as a treatment component of dysphagia in stroke patients. . . . The systematic performance of different exercises and maneuvers, provided for in an individualized therapeutic plan and adjusted to the clinical condition, is recommended in stroke patients with dysphagia."

The American Speech Language Hearing Association (ASHA, 2022) stated that "Patients who are tactically defensive may need approaches that reduce the level of sensory input initially, with incremental increases as tolerance improves. Sensory stimulation may be useful for those with reduced response, overactive response, or limited opportunity for sensory experience."

Mirror Therapy

He et al (2022) stated that post-stroke dysphagia is a common symptom following stroke and one of the most frequent and severe complications of stroke. In the past several years, mirror therapy has generated significant research interest as a non-invasive therapeutic and rehabilitative intervention for the treatment of post-stroke dysphagia and has been examined in several single-center RCTs. In a systematic review and meta-analysis, these investigators examined the safety and effectiveness of mirror therapy for the treatment of patients with post-stroke dysphagia. A total of 7 databases were searched from inception to the December 31, 2021, including PubMed, the Cochrane Library, Web of Science, China National Knowledge Infrastructure (CNKI), Chinese Biomedical Literature Service System (SinoMed), Wan-fang database, and the Chinese Scientific Journals Database (VIP) from inception to December 31, 2021. The primary outcome measure was effectiveness, as measured by clinical effectiveness rate. Secondary outcomes included the water swallowing test and the incidence of pneumonia. Furthermore, these researchers employed the Cochrane Risk of Bias Tool to examine the risk of bias. Potential publication bias was evaluated by applying Egger's bias indicator test and by assessing the symmetry of data when visualized as funnel plots. A total of 5 RCTs (135 subjects in the experimental group and control group) were found to report the use of mirror therapy for post-stroke dysphagia and were included in this study. No publication bias was detected. Meta-analysis showed that mirror therapy had a positive effect on the rate of clinical effectiveness [OR = 4.22; 95 % CI: 2.3 to 7.73] and the water swallowing test (MD = -0.76; 95 % CI: -1.29 to -0.22]. Moreover, mirror therapy reduced the incidence of pneumonia (OR = 0.13; 95 % CI: 0.03 to 0.49). Subgroup analyses indicated that mirror therapy during the acute phase was robust but was unstable during the convalescent phase. Sensitivity analysis revealed that the results generated by this meta-analysis were robust and stable. The authors concluded that available evidence appeared to suggest that mirror therapy may have a role in the management of post-stroke dysphagia but has yet to be fully confirmed. These researchers stated that existing evidence from clinical trials suggested that evidence relating to the safety of mirror therapy for patients with post-stroke dysphagia is insufficient. They stated that there is a need to perform further studies with a larger sample size using more objective indicators.

The authors stated that this study had several drawbacks. First, the currently available evidence related to the use of mirror therapy for post-stroke dysphagia originated from a relatively small sample size. Second, there were limited numbers of subjects in each trial and limited analysis of outcome data. Third, safety indicators may not be optimal and follow-up data are lacking; thus, these investigators could not fully examine the safety profile of mirror therapy or the long-term effectiveness of this technique in the target patient population. Fourth, although the authors carried out specific searches in a wide range of databases, all of the included trials were conducted in China; as such, these findings may not be representative on a wider scale.

Neuromuscular Electrical Stimulation

In a meta-analysis, Carnaby-Mann and Crary (2007) evaluated the effect of transcutaneous neuromuscular electrical stimulation (NMES) on swallowing rehabilitation. The authors concluded that this preliminary meta-analysis revealed a small but significant summary effect size for transcutaneous NMES for swallowing. Because of the small number of studies and low methodological grading for these studies, caution should be taken in interpreting this finding. These results support the need for more rigorous research in this area. This is in agreement with the observation of Steel et al (2007) who noted that although ES approaches to the restoration and rehabilitation of swallowing is an exciting area of research which holds promise for future clinically relevant technology and/or therapy, implementation of ES in clinical swallowing rehabilitation settings still remains pre-mature.

Miller et al (2022) stated that NMES has been used as a therapeutic option in the treatment of dysphagia for several years. In a previous review of the literature, it was concluded that NMES might be a valuable adjunct in patients with dysphagia and in

patients with vocal fold paresis. However, due to different stimulation protocols, electrode positioning and various underlying pathological conditions, it was difficult to compare the studies that were identified; and it was concluded that more empirical data is needed to fully understand the benefits provided by NMES. In a systematic review, these researchers assessed recent studies regarding a potential effectiveness of transcutaneous NMES applied to the anterior neck as a treatment for dysphagia considering these different aspects. For this systematic review, a selective literature research in PubMed was performed on May 5, 2021 using the terms electrical stimulation AND dysphagia and screened for inclusion criteria by 2 reviewers in Rayyan. The search resulted in 62 hits. Studies were excluded due to their publication language; because they did not meet inclusion criteria; because the topical focus was a different one; or because they did not qualify as level 2 studies. A total of 18 studies were identified with varying patient groups, stimulation protocols, electrode placement and therapy settings. However, 16 studies have reported of beneficial outcomes in relation with NMES. The authors concluded that there is a considerable amount of level 2 studies that suggested that NMES is an effective therapeutic option, especially when combined with TDT for patients with dysphagia following stroke and patients with Parkinson's disease, or with different kinds of brain injuries. Moreover, these researchers stated that further investigation is needed to clarify which stimulation protocols, parameters and therapy settings are most beneficial for certain patient groups and degrees of impairment. They also noted that in this systematic review, data pooling and statistical analysis could not be performed due to the inhomogeneity of study protocols.

Peripheral Magnetic Stimulation

In a pilot study, Momosaki et al (2015) examined the safety and feasibility of a 6-day protocol of in-hospital repetitive peripheral magnetic stimulation combined with intensive swallowing rehabilitation (rPMS-ISR) for post-stroke dysphagia. Subjects were 8 patients with dysphagia caused by bilateral cerebral infarction (age of 62 to 70 years; time from onset of stroke: 27 to 39 months). Repetitive PMS was applied to the suprahyoid muscles, at strength set at 90 % of the minimal intensity that elicited pain with a parabolic coil. One train of stimuli comprised 20 Hz for 3 sec followed by 27-sec rest. A single session included delivery of repetitive 20 trains of stimuli over 10 mins, followed by 20 mins of swallowing rehabilitation. Each patient received this combination treatment twice-daily, morning and afternoon, over 6 consecutive days. Swallowing function was evaluated before and after intervention. Repetitive PMS-ISR induced significant improvement in swallowing ability, laryngeal elevation delay time, penetration aspiration scale, and swallowing quality of life ($p < 0.01$), but had no significant effect on the functional oral intake scale. The authors concluded that the 6-day in-hospital RPMS-ISR protocol appeared safe and feasible for post-stroke patients with dysphagia. The combination protocol improved swallowing function. They stated that further larger studies are needed to confirm its effectiveness.

Hwang et al (2022) stated that recently, a therapeutic method to stimulate the suprahyoid muscle using PMS for dysphagia rehabilitation has been reported; however, clinical evidence, application protocol, and intervention method remain unclear. In a systematic review, these investigators evaluated clinical studies of PMS used for rehabilitation of dysphagia. They carried out a literature search of 4 databases (Medline, Embase, CINAHL, and Web of Science) to identify relevant studies published on the use of repetitive PMS (rPMS) for swallowing-related muscles between 2010 and 2022. A total of 7 studies were reviewed; RCTs and 1-group pre-post, case study designs were included. In the included studies, rPMS was employed to strengthen the submental suprahyoid muscles. The intervention regimen varied. The rPMS was used at a frequency of 30 Hz for 2 s. Rest time ranged from 8 s to 27-28 s. The number of intervention sessions ranged from 2-3 to 30. The intensity ranged from pain-inducing minimum intensity (90 % of maximum stimulus output) to non-painful intensity (70% to 80 % of maximum intensity). The rPMS on the suprahyoid muscles had positive effects on physiological changes in the swallowing function, such as displacement of the hyoid bone, muscle strength (cervical flexor, jaw-opening force), swallowing safety, swallowing performance, and swallowing-related QOL. Participants also reported little pain and adverse reactions during rPMS. The authors concluded that although rPMS is a therapeutic option that can help improve the swallowing function as a non-invasive stimulation method in the rehabilitation of dysphagia, clinical evidence is needed for the development of clear stimulation protocols and guidelines. Moreover, these researchers stated that there are issues to be considered when applying PMS to patients with dysphagia. First, since the swallowing muscle is a very small muscle located in a narrow space under the hyoid bone and chin, accurate

stimulation focusing is important. However, the PMS equipment currently used is not a device developed for stimulating the swallowing muscle; thus, it is not a dedicated device considering the anatomical structure and characteristics of the swallowing muscle. Furthermore, the protocol for optimal stimulation is still unclear. As a consequence, clearer evidence is needed via coil design, which reflects the structure and characteristics of the swallowing muscle and various stimulation protocols for the effective rehabilitation of dysphagia.

Hyperbaric Oxygen Therapy for the Treatment of Radiation-Induced Dysphagia

In a systematic review, Philipsen et al (2024) examined the role of hyperbaric oxygen therapy (HBOT) in patients with dysphagia following radiation therapy (RT) for head and neck cancer (HNC). These investigators carried out a systematic search in the electronic databases Ovid Medline, Ovid Embase, and Cochrane Central Register of Controlled Trials for relevant studies until March 14, 2023. No restriction on language or publication date. The criteria for inclusion: patients with HNC who had received both RT and HBOT as a preventive treatment against swallowing difficulties; to preserve swallowing function, or to promote swallowing difficulties. These researchers identified 1,396 records. After removal of 31 duplicates, 1,365 records were accessible for title and abstract screening yielding 53 studies for full-text assessment. A total of 6 studies met the eligibility criteria and were included for qualitative analysis. The authors concluded that evidence of HBOT benefits in patients with dysphagia following RT for HNC was inconsistent. These researchers stated that well-designed clinical trials using validated outcome measures and long-term follow-up are needed.

Artificial Intelligence in the Management of Dysphagia

Gallano et al (2025) noted that artificial Intelligence (AI) is transforming speech-language pathology (SLP) and dysphagia management, offering innovative solutions for assessment, diagnosis, and rehabilitation. These investigators examined AI applications in these fields from 2014 to 2024, with an emphasis on implementation challenges in Latin America. They analyzed key AI technologies --including deep learning (DL), machine learning (ML) algorithms, as well as natural language processing -- that have shown high accuracy in detecting voice disorders, analyzing swallowing function, and supporting personalized rehabilitation. These researchers identified 3 primary domains of AI application: diagnostic tools with improved sensitivity for speech-language disorders, rehabilitation technologies that enable customized therapy, and tele-health platforms that expand access to specialized care in under-served regions. However, significant barriers persist, especially in Latin America, where limited infra-structure, insufficient linguistic adaptation, and scarce regional data-sets hamper widespread implementation. In a pilot study, these investigators examined commercially available large language models for rehabilitation planning demonstrated their potential utility in generating structured therapy activities, especially in resource-constrained settings. The authors concluded that while AI shows promise in enhancing clinical work-flows and expanding service delivery, the evidence suggests that current applications remain predominantly focused on diagnosis rather than integrated rehabilitation. These researchers highlighted the need for culturally and linguistically adapted AI models, expanded regional research collaborations, as well as regulatory frame-works that ensure ethical AI integration into SLP and dysphagia care, positioning these technologies as complementary tools that enhance rather than replace clinical expertise.

Furthermore, UpToDate reviews on “Oropharyngeal dysphagia: Clinical features, diagnosis, and management” (Lembo, 2025), and “Approach to the evaluation of dysphagia in adults” (Fass, 2025) do not mention artificial Intelligence as a management tool.

Transoral Holmium YAG Laser Resection for the Treatment of Dysphagia

In a single-case report, Teasdale et al (2025) described a pioneering approach to the treatment of superior thyroid cornu syndrome (STCS), a condition characterized by dysphagia and globus sensation due to anatomical variations in the thyroid cartilage, often resulting in significant patient discomfort and mis-diagnosis. This case entailed a man in his 50s, presenting with a persistent sensation of a lump in his throat and difficulty swallowing without associated pain or systemic symptoms, underwent a comprehensive diagnostic work-up that included flexible naso-endoscopy, MRI, contrast swallow, and CT scans, showing a medially displaced superior cornu of the thyroid cartilage. Traditional surgical interventions offered limited precision and potential

for soft tissue trauma. This report introduced the use of the holmium:yttrium-aluminum-garnet (Ho:YAG) laser for transoral resection of the affected cartilage, a technique more commonly associated with urological and salivary stone treatments. This innovative application resulted in successful symptom resolution without significant complications, highlighting the Ho:YAG laser's safety and effectiveness for STCS treatment and encouraging further investigations of its use in otolaryngology.

Furthermore, an UpToDate review on "Oropharyngeal dysphagia: Clinical features, diagnosis, and management" (Lembo, 2025) does not mention laser as a therapeutic option.

Per Oral Plication of the Esophagus (POPE)

Hedberg et al (2023) noted that endoscopic sleeve gastropasty is an example of endoscopic sutured plications being used to remodel a gastro-intestinal (GI) organ. With per-oral plication of the esophagus (POPE), similar plications are used to remodel the dilated and redundant mega-esophagus of end-stage achalasia. Redundancies and dilations can also develop in the neo-esophagus of a patient with prior esophagectomy. Mega-esophagus and a redundant neo-esophagus can both lead to debilitating dysphagia, regurgitation, and recurrent aspiration pneumonia. Traditionally, this anatomic problem requires complex revisional or excisional surgery, to which POPE offers an incisionless alternative. This article was a dynamic manuscript with video demonstration of POPE, as well as review of 5 cases performed in 1 year. Data were collected in a prospectively maintained data-base, and the institutional review board (IRB)-approved retrospective review for this publication. The procedure was carried out using a dual-channel upper endoscope fixed with an endoscopic suturing device, with the patient in a supine position under general anesthesia. POPE was technically completed in all cases with no serious complications, and patients either went home the same day or spent 1 night for observation. Most patients reported immediate and substantial symptomatic improvement. Objective pre- and post-measures included esophagram and nuclear gastric emptying studies. The authors discussed early experience at 1 center with POPE, with detailed description of the procedure and technical considerations. An accompanying video reviewed 2 cases, 1 with mega-esophagus and 1 with a gastric conduit. These researchers stated that while this novel procedure has limited and rare indications, it offers a low-morbidity solution to a challenging anatomic problem that traditionally requires invasive surgery.

The authors stated that this study had several drawbacks. First, this was a very limited case series (n = 5), intended as a descriptive study of the POPE procedure and its indications. Second, these investigators could draw no conclusions regarding superiority of this approach to end-stage achalasia/neo-esophageal anatomic issues over traditional esophagectomy or surgical conduit revision. While they presented some objective data showing improvement of emptying following POPE, there was no standardized pre- and post-procedure evaluation for these 5 patients. Third, given the subjective nature of symptomatic assessment and the execution of the procedure itself, standardized pre- and post-operative assessment by timed barium esophagography would be valuable in further examination of this technique.

Crafts et al (2024) stated that POPE is an endoscopic procedure used to improve emptying of the dysfunctionalized esophagus or gastric conduit, with the hope of improving symptoms and quality of life (QOL). As this procedure has only been carried out in the U.S. for the past 4 years, safety and effectiveness have not been well-established. In a retrospective, case-series, single-center study, these researchers examined the effectiveness of POPE for the treatment of impaired emptying in achalasia and post-esophagectomy patients. This trial included patients who underwent POPE between 2019 and 2023. Data collected included demographics, pre-operative diagnoses and treatments, imaging, endoscopic data, operative intervention, 90-day complications, and response to treatment. QOL and patient satisfaction data were collected by phone survey. A total of 17 cases were identified, encompassing 13 primary procedures and 4 repeat POPEs (re-POPE). A total of 8 patients had end-stage achalasia and 5 had impaired gastric emptying following esophagectomy with gastric conduits. Median age of this cohort was 65 years and median ASA was 3, with 38.5 % women. POPE was carried out with 2 to 6 plication sutures in an average of 75 mins. The majority of patients were discharged home on the same day. For the 17 procedures, there were 4 complications - 2 patients required antibiotics for pneumonia, while 4 required procedural intervention. There were no deaths. Pre-operative symptoms improved or resolved at initial follow-up in 82.3 % of patients; 4 patients experienced symptom recurrence and required re-POPE, 1 with achalasia and 3 with gastric conduits. Although all achalasia patients had an "end-stage

esophagus", none had required esophagectomy since the introduction of POPE. The authors concluded that POPE is an endoscopic procedure that was effective in relieving emptying difficulties for the end-stage esophagus and gastric conduit. It may obviate the need for esophagectomy or conduit replacement. Furthermore, it can be repeated in select patients. Moreover, these researchers noted that while the risk profile of complications was favorable compared to alternative operations, patients with gastric conduits were at higher risk.

Uchima et al (2025) noted that delayed gastric conduit emptying (DGCE) affects 15 % to 39 % of patients following esophagectomy, increasing the risk of aspiration, pneumonia, as well as malnutrition. "Sump" formation with chronic dilation often necessitates surgical intervention. POPE is a novel endoscopic technique adapted from endoscopic sleeve gastropasty (ESG) to re-model the neo-esophagus (and end-stage achalasia mega-esophagus) and improve emptying 2, with 82.3 % symptom improvement, although 23.5 % requiring repeat intervention. Studies of ESG have shown that 83.64 % of sutures persist at 12 months, with 70.9 % maintaining adequate tension. Combining mucosal denudation techniques like argon plasma coagulation (APC) or endoscopic mucosal resection (EMR) may enhance submucosa to submucosa apposition resulting in better suture retention and durability. These investigators presented a modified POPE by combining mucosal denudation techniques, to address sump-related retention and improve gastric emptying in a patient with refractory DGCE. This case entailed a 72-year-old woman with esophageal adenocarcinoma who underwent partial esophagectomy with gastric conduit reconstruction in 2019. She developed early satiety, worsening heart-burn when supine, and progressive oral intolerance, necessitating total parenteral nutrition. Imaging confirmed a neo-esophageal sump with significant retention. Dietary adjustments, prokinetics, and pyloric BTX injection failed, with worsening symptoms and bile reflux. A multi-disciplinary team recommended POPE for sump re-modeling. Modified POPE was carried out under general anesthesia in the supine position, using a suturing device without an over-tube, combining with EMR and APC. Endoscopic and radiologic assessment confirmed improved conduit alignment and pyloric passage. The patient was discharged after 48 hours without complications and tolerating diet. At a 4-month follow-up, endoscopy confirmed sustained re-modeling with symptom improvement and no retention. The authors concluded that modified POPE was a feasible, minimally invasive technique for refractory DGCE due to sump formation. Moreover, these researchers stated that further investigations are needed to evaluate long-term effectiveness.

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